



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

Luminova : Towards Sustainable Light-bulbs

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Abstract

This case study presents LumiNova, a modular and repairable LED light bulb designed as a circular alternative to conventional products on the market. Today, most LED bulbs are permanently glued together, making repair impossible and forcing users to throw away the whole product even when only one part has failed. This leads to large amounts of electronic waste and poor material recovery at end of life. To address this, LumiNova was redesigned around three key innovations: a clip-based assembly that allows tool-free disassembly, a built-in circuit that identifies which part has failed, and lower-impact materials including a biodegradable PCB substrate and a gold-free surface finish. A full life cycle assessment, material flow analysis, and circularity evaluation were carried out to measure the environmental benefits of this approach. The results show a reduction in carbon footprint from 7.21 to 5.01 kg CO₂ eq. over the product lifetime, a 36% decrease in production-related impacts, and an increase in the circularity index from 24% to 71%, mainly due to the longer product lifetime enabled by modularity. Two new dedicated recovery streams for the PCB and the aluminium cap are also created at end of life. Overall, this work shows that modular design combined with lower-impact materials and a redesigned supply chain is a practical and scalable path toward more circular consumer electronics.

1 Introduction

The transition towards a circular economy has become a major challenge for electronic equipment. Although LED light bulbs are among the most energy-efficient lighting technologies available today, their design remains largely based on a linear model of production, use, and disposal (see Figure 1). With approximately two billion light bulbs sold annually in Europe, the rapid adoption of LED technology is generating an increasing stream of waste electrical and electronic equipment (WEEE) and raising concerns regarding resource consumption and material recovery (*European Commission, 2026b; Signify, 2025a*).

Modern LED bulbs are complex multi-material products composed of polymers, metals, electronic components, and printed circuit boards (PCBs). Most commercial products are permanently assembled using adhesives and sealed structures, making repair and disassembly difficult. Consequently, the failure of a single component often leads to the disposal of the entire bulb, despite most components remaining functional. Previous studies have shown that such designs limit repair, reuse, and refurbishment strategies while reducing the recovery of valuable materials such as aluminium, copper, gold, and rare earth elements (*Cenci et al., 2020; Illés & Kékesi, 2023; Martins et al., 2020; Rahman et al., 2021*).



Figure 1: Current LED bulb end-of-life management. Most bulbs are shredded and processed as mixed waste streams, limiting the recovery of valuable materials.

The objective of this work is to develop and evaluate *LumiNova*, a modular and repairable LED bulb designed according to circular economy principles. The proposed concept combines a disassemblable architecture, an integrated fault-detection system, and lower-impact material alternatives to reduce environmental burdens throughout the product life cycle.

Three hypotheses are investigated. First, modularity can extend product lifetime by enabling component-level replacement instead of full product disposal. Second, the use of recycled and lower-impact materials can reduce production-related environmental impacts. Third, dedicated recovery streams enabled by disassembly can improve material circularity and end-of-life resource recovery.

The significance of this work extends beyond the lighting sector. As European regulations increasingly promote reparability and circular product design, LED bulbs provide a representative case study for exploring how consumer electronics can transition from a disposable model towards a circular system. The methodology developed here may therefore serve as a framework for other electronic products facing similar challenges.

The remainder of this report is organized as follows. Section 2 reviews the state of the art in LED recycling and circular design. Section 3 presents the *LumiNova* concept and evaluates its environmental, material, and economic performance through material flow analysis, life cycle assessment, circularity metrics, and cost modelling. Finally, recommendations and future implementation pathways are discussed before concluding the study.

2 State of the Art

Decarbonisation of light bulbs is the subject of recent research from the biggest lighting industries. A first lever is the improvement of energy efficiency during the use phase. Signify has therefore developed its Philips UltraEfficient range, whose products are presented as at least 40% more energy-efficient than standard Philips LED bulbs, while some company documents indicate improvements of up to 50-60% compared with first-generation LED products (*Signify, 2024, 2025e*). Another major lever is lifetime extension. By increasing durability, fewer bulbs need to be produced, transported and treated at the end of life over the same lighting service period. For example, Signify reports nominal lifetimes of up to 50 000 hours for Philips MASTER

UltraEfficient LED bulbs, which helps reduce replacement frequency and associated material flows (*Signify, 2025b*).

Furthermore, it is clear that lighting industries are trying to integrate more recycled materials, especially in plastic and aluminium parts. This strategy reduces the demand for virgin raw materials and therefore lowers the embodied carbon associated with material extraction and production. For instance, LEDVANCE has developed its NATURELOOP range, in which the plastic components of certain lamps contain at least 40% post-consumer recycled plastic (*LEDVANCE, 2023*). Similarly, Signify recently introduced Philips LED tubes made with an average of 40% post-consumer recycled plastic in their plastic content, while also using packaging made from 80% recycled paper (*Signify, 2025d*). Although this approach does not fully eliminate the environmental impact of lighting products, it represents a practical industrial pathway to reduce reliance on virgin materials and improve the circularity of product manufacturing.

In addition, Signify has developed other strategies to decarbonize light bulbs in which additive manufacturing plays an important role. According to the European Circular Economy Stakeholder Platform, Signify’s 3D-printed light bulbs can achieve up to 47% lower carbon footprint than conventionally manufactured ones (*European Circular Economy Stakeholder Platform, 2021*). This reduction is mainly explained by lower material consumption, fewer production steps, lighter products, and the possibility of local and on-demand manufacturing, which also reduces transport-related emissions. In addition, Signify highlights that these light bulbs are made from recyclable polycarbonate and can incorporate recycled feedstock, such as secondary plastics, thereby reducing dependence on virgin raw materials and supporting closed material loops. The design freedom offered by 3D printing also allows a reduction of the number of parts and adhesives, which can simplify product architecture and improve circularity at the end of life. However, these advantages must be interpreted with caution. First, most of the reported benefits currently concern light bulb housings rather than the electronic core of lighting products, such as LED chips, drivers or printed circuit boards, which remain major environmental hotspots. Second, despite its flexibility and sustainability potential, 3D printing still faces important industrial limitations. Compared with conventional high-throughput processes such as injection moulding or metal forming, additive manufacturing is generally slower and less economically competitive for very large production volumes. Signify has expanded this technology in several facilities, showing that industrial deployment is possible, but it is still better suited to customised applications than to fully replacing the actual manufacturing in the near term. As a result, 3D printing should not be seen as a universal solution for the lighting industry, but more as a promising complementary pathway for reducing embodied carbon, and enabling recycled material use.

Moreover, it should be noted that another way to decarbonise light bulbs used in industry is the improvement of product packaging. Although packaging contributes less to the overall footprint than the use phase or electronic components, it remains important for high-volume consumer products such as light bulbs. Current industrial strategies focus on replacing fossil-fuel-based plastic packaging with recycled and recyclable paper-based alternatives, while also reducing packaging size and weight to lower transport-related emissions. For example, Signify states that its packaging policy requires the use of 80% recycled paper and reports that the phase-out of plastic from consumer packaging avoids more than 2 500 tonnes of plastic per year while reducing transport and material-related carbon emissions by around 6 000 tonnes of CO₂ annually (*Signify, 2020, 2025c*). Similarly, LEDVANCE reports that the packaging of its NATURELOOP lamps is plastic-free, made from at least 80% recycled paper, and fully recyclable (*LEDVANCE, 2024*). This shows that packaging is also an active part of lighting product decarbonisation strategies rather than a secondary consideration.

The transition toward a circular economy in the lighting industry is currently hindered by the complex, multi-material nature of modern LED lamps, which makes traditional bulk material recovery inefficient (*Rahman et al., 2021*). Current research emphasizes shifting from simple recycling (R8) toward higher-order 10R strategies (Fig. 2) such as reuse (R3), repair (R4), and refurbishment (R5) to maximize value retention and extend product lifespans (*Rahman et al., 2021*).

Technical experts propose ”Design for Recycling (DfR)” rules, specifically urging that parts be readily separable and that modular component structures be adopted (*Rahman et al., 2021*). To make modularity effective, research highlights ten specific DfR rules (*Rahman et al., 2021*) (Fig. 3). Rule 8 specifies that parts must be readily separable, suggesting a shift toward modular and separable component structures rather than glued or permanent assemblies. Rule 10 emphasizes designing for the easy liberation of materials, which prevents the contamination of high-value fractions (like aluminium) with minor electronic elements (*Rahman et al., 2021*).



Figure 2: 10 R's of circular economy (M. & Y., 2026)

- 1 Oversimplification focusing on general recycling rules would not produce the intended goal of resource efficiency (RE)
- 2 Requires model and simulation-based quantification
- 3 Design data should be available and compatible
- 4 Appropriate technology infrastructure and relevant tools
- 5 Linking design tools with recycling process simulation for designers to be informed about the accurate picture of metal and material recycling
- 6 Identify and minimize the use of materials that contaminate the recycling process
- 7 Identify and minimize the use of components that make recycling difficult for combined and applied materials
- 8 Design important; parts to be readily separable
- 9 Labeling based on product or component recovery
- 10 Designed for easy liberation of materials

Figure 3: 10 Rules for DfR (Rahman et al., 2021)

Currently, the "useful life" of an LED lamp often ends not at total failure, but when luminous efficacy declines to 70% (Rahman et al., 2021). Poor heat dissipation or low-quality electronics frequently cause early failure of the entire unit (Rahman et al., 2021). A modular approach addresses this by allowing for the replacement of specific defective parts, such as the PCB or LED chip, while retaining stable components like the heat sink (Rahman et al., 2021). Implementing "Lighting as a Service" (LaaS) models is also discussed as a way to track and upgrade obsolete products, informing designs that promote reuse rather than disposal (Rahman et al., 2021).

The Lighting as a Service (LaaS) model supports modular design by shifting ownership of the lighting equipment from the consumer to the manufacturer (Rahman et al., 2021). This shift in ownership directly incentivizes producers to adopt modular structures to manage the product throughout its entire life cycle more efficiently (Rahman et al., 2021). Because the manufacturer retains responsibility for the bulb's performance, they are motivated to design products where defective parts can be easily identified and replaced (Rahman et al., 2021). Modular design allows for circular strategies which implement repair (R4) and refurbishment (R5), without the cost of replacing the entire unit (Rahman et al., 2021). Under a LaaS model, modularity allows a manufacturer to upgrade specific components (like the LED chip) to gain efficiency benefits while keeping stable components (like the housing or heat sink), making frequent upgrades economically viable (Rahman et al., 2021). LaaS allows a "dedicated entity" (the manufacturer) to track a product's performance data. This information helps them make informed decisions on when to upgrade or replace specific modules before they reach the end-of-life (EoL) stage. Retaining ownership allows producers to collect failed units and use the data to inform future modular designs that promote the reuse of components.

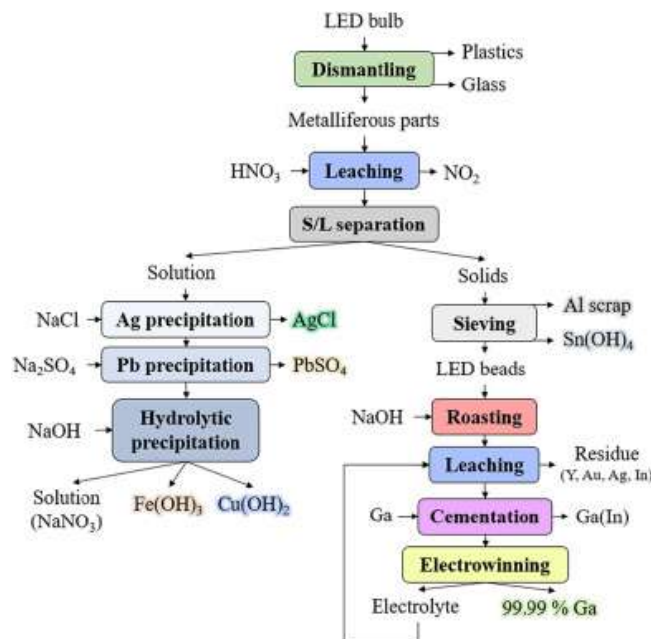


Figure 4: The flowsheet for the proposed recycling method of waste LED bulbs. (Illés & Kékesi, 2023)

Effective material recovery requires the clean separation of distinct modules. Studies have demonstrated that mechanical operations, including comminution, sieving, and magnetic/electrostatic separation, can achieve 98–100% separation efficiency for major components like the polymeric cover, metallic housing, Edison screw, and PCB (Martins *et al.*, 2020). Aluminum (from housings) and PCBs represent significant economic value. For instance, the PCB alone can account for over 50% of a lamp’s total recovery value (Martins *et al.*, 2020; Rahman *et al.*, 2021). Once components are separated, hydrometallurgical processes (leaching) can selectively recover base metals like Copper, Tin, and Lead, as well as critical rare metals like Gallium and Indium from the LED chips (Flowsheet illustrated in Figure 4) (Illés & Kékesi, 2023). Passivating aluminum during chemical leaching allows it to be recovered as a clean, recyclable scrap (Illés & Kékesi, 2023).

While manual disassembly is vital for producing intact components for reuse, it is often seen as too labor-intensive for developed economies (Rahman *et al.*, 2021). Research suggests integrated management frameworks that combine robotic or mechanical disassembly with advanced sorting (Rahman *et al.*, 2021).

Recent research by Cenci *et al.* has highlighted that understanding how materials are distributed across different parts is essential for an efficient recycling route (see Figure 5) (Cenci *et al.*, 2020). They divided the lamps into four main categories: carcass, LEDs, printed circuit board (PCB), and LED module. A major finding of this study is that although LEDs have a higher concentration of gallium, the total quantity of gallium is actually larger in the PCBs (accounting for over 79–85% of the total gallium content) (Cenci *et al.*, 2020). This suggests that a modular design allowing for the easy recovery of the PCB is vital for capturing this critical metal. The carcasses are primarily composed of pure polymers (like Polycarbonate) and aluminium. The paper suggests that separating the carcass as the first step of recycling avoids diluting high-value electronic fractions and allows these materials to be recovered in their pure form, which is more energy-efficient (Cenci *et al.*, 2020).

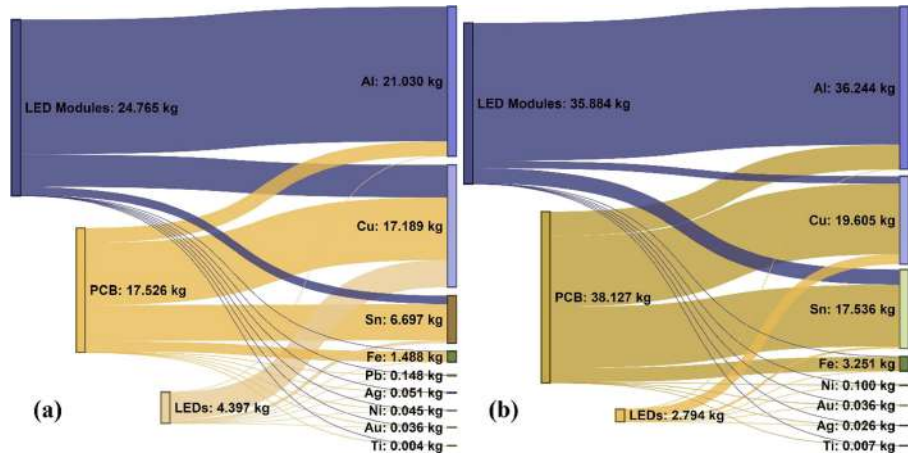


Figure 5: Sankey diagrams of the recovered materials from tubular (a) and bulb (b) lamps components (Cenci *et al.*, 2020)

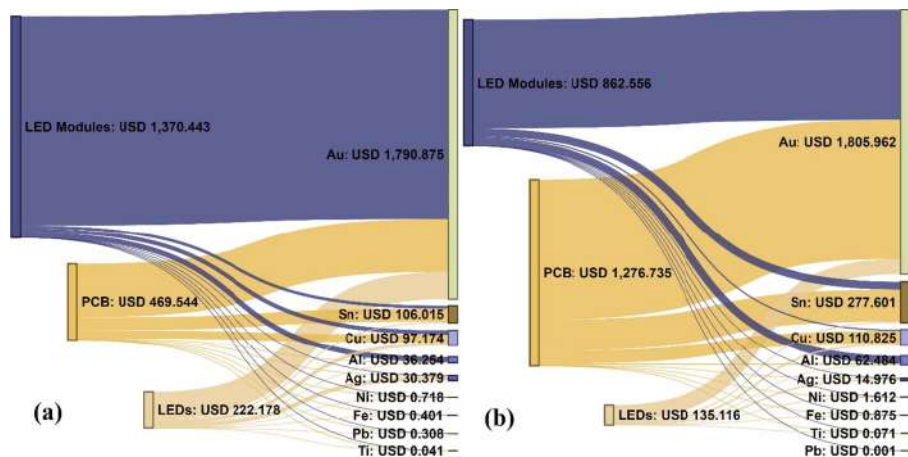


Figure 6: Sankey diagrams of the recovered economic values from tubular (a) and bulb (b) lamps components (Cenci *et al.*, 2020)

Cenci *et al.* estimate the total economic recovery potential to be approximately USD 2,406 per ton for tubular lamps and USD 2,595 per ton for bulb lamps (Cenci *et al.*, 2020). While the LEDs have the smallest mass fraction, they are the most valuable component per kilogram (valued at roughly USD 20/kg) due to their high concentration of gold and silver (Cenci *et al.*, 2020). Gold is identified as the most valuable single material, accounting for nearly 80% of the total economic recovery value of the lamps (Figure 6) (Cenci *et al.*, 2020).

Cenci *et al.* note that current bulk recycling methods (which do not use component segregation) lose critical materials like gallium and rare earth elements. They argue that eco-design strategies to facilitate dismantling are essential to maximize resource recovery (Cenci *et al.*, 2020). They suggest a sequence of milling followed by electrostatic separation for LEDs specifically (Cenci *et al.*, 2020). This would separate conductive metals (Au, Ag, Cu) from non-conductive materials (Ga and rare earths), but this is only possible if the LEDs are first separated from the rest of the unit (Cenci *et al.*, 2020). The paper found lead in some components of tubular lamps, highlighting that modularity could also help in isolating and safely managing hazardous substances (Cenci *et al.*, 2020).

3 Case Study Results and Discussion

3.1 Define Circular Product

3.1.1 Product Definition, Value Proposition, and Market Position

A LED light bulb is an electrical lighting device that converts electricity into light using light-emitting diodes (LEDs). Compared to traditional incandescent or halogen bulbs, LED bulbs consume less energy, generate less heat, and have a much longer lifetime. They are commonly used in households, offices, and public buildings for general lighting applications.

A standard LED light bulb has one main purpose: providing efficient, safe, and long-lasting lighting. The primary user need is to have access to a reliable, energy-efficient, and affordable light source that requires minimal maintenance over its lifetime.

The LED lighting market is today dominated by a few large companies, including Signify (formerly Philips Lighting), OSRAM, Ledvance, and GE Lighting, all offering products that follow a similar approach: high-volume, low-cost bulbs designed to be replaced rather than repaired (Signify, 2025a). The end-of-life treatment of these products is largely linear, with most bulbs being shredded as a single unit at recycling facilities, leading to significant material losses (United Nations Environment Programme, 2017).

LumiNova’s product meets all the same functional requirements as these conventional alternatives, but the innovation is not about light output or energy efficiency. As shown in Figure 7, a LED bulb is made of several separate parts: diffuser, LED module, PCB, heat sink, and base, and it is this structure that LumiNova uses to make the product more circular and less carbon-intensive, directly tackling the limits of current industry practice.



Figure 7: Disassembled LED bulb, adapted from (Earth911, 2024)

The main observation behind LumiNova’s design is that a large proportion of LED bulbs is discarded because of a single component failure, most often the driver PCB, and sometimes the LED board (Prakash *et al.*, 2016).

Instead of replacing the whole bulb, LumiNova proposes a modular architecture where each part (LED board, driver PCB, and housing) can be removed and replaced by the user without any tools or technical knowledge, as detailed in the Business Model Canvas (see Appendix S1).

This repairability relies on two key material changes presented in the Safe & Circular Product Redesign Workshop (see Appendix S2): a glue-free, clip-based assembly replacing conventional adhesive bonding, and a biodegradable resin PCB substrate replacing conventional epoxy (FR4). These choices and their implications are discussed in more detail in the following sections.

These changes therefore represent an important shift in light bulb design, moving away from a disposable product towards a modular and repairable one. Among the dominant market players, Signify was chosen as the reference company for this case study, being one of the most well-known companies in the sector with the largest market share (*Signify, 2025a*). Its product line is representative of current industry practice, with conventional non-disassemblable designs that offer no component-level repairability or end-of-life material recovery, making it a relevant benchmark to evaluate the potential impact of LumiNova’s modular approach.

3.1.2 Current sustainability and decarbonization strategies of sector

The decarbonization strategies of actual light bulb products differ depending on the region of the world. The European Union appears to have one of the most stringent regulations regarding the recycling of these products (*United States Environmental Protection Agency, 2026*) (*Ministry of Economy, Trade and Industry, Tokyo, Japan, 2010*). Indeed, they are under the common norm of Waste from Electrical and Electronic Equipment (WEEE), that organizes the separate collection and recovery of materials (*European Commission, 2026b*). In practice, the process is done by the following way :

- collection of light bulbs is done in shop or directly in sorting centers
- products are brought in centers to be shredded
- components are separated (metals, polymers, electronic compounds)
- parts are recycled or eliminated.

After shredding, specific techniques are used to retrieve the different materials, such as magnetic or water separation (*RTS, 2023*). However, the collected materials are not directly used for new light bulb products, but they are sent to recycling facilities, where they can be used as secondary raw materials. Less than 20% of materials in the product follow this path, as the rest is eliminated (*European Commission, 2026b*). Specifically, even if separation techniques exist, it is hard to recover all materials in the bulb separately after the shredding process. Those complex parts are a mixture of materials glued together (like polymers contaminated with resin), electronic parts, or Rare Earth Elements that cannot be recycled as there are no large scale recycling facilities due to their low amount in electronic products.

It should be noted that, every country in the European Union follows this norm but implements and organizes it in its own way (with its own sorting centers, fundings, ...). And in the case of Switzerland, they do not directly apply the WEEE as they are not part of UE but the general principle is similar.

3.1.3 Next-Generation Circular Product Design

In response to these limitations, LumiNova proposes a competitive circular alternative built around three interconnected innovations: a modular mechanical design, a built-in failure detection system, and lower-impact material substitutions. The core enabling technology is a snap-fit neck system that replaces permanent adhesive bonding with mechanical clips, allowing tool-free disassembly of the three main modules (LED board, driver PCB, and housing) without any technical knowledge. This is a direct response to current industry practice, where bulbs are permanently sealed, making disassembly nearly impossible and forcing disposal of the entire product even when only one part has failed (*Prakash et al., 2016*). A built-in failure detection circuit complements this modular design by identifying which of the two possible failure points has broken down before the user opens the bulb, ensuring that only the defective module is replaced and reducing unnecessary electronic waste over the product lifetime. On the materials side, two important substitutions are introduced. The conventional epoxy-based FR4 PCB substrate is replaced with a biodegradable resin, which simplifies component separation and reduces the toxicity of the product at end of life. In parallel, the standard gold-based ENIG surface finish is replaced with an OSP finish, removing gold and nickel from the board entirely. Despite being

present in very small quantities, gold carries an extremely high carbon footprint and represents one of the more environmentally harmful materials per gramme in a conventional LED bulb. Together, these technologies enable a new supply chain logic. Rather than replacing a whole bulb upon failure, customers can identify the faulty module and purchase only that part through hardware stores or online channels, at a much lower cost than buying a complete new bulb. Spent modules are then returned through retail collection points under a take-back program, creating two new dedicated recovery streams at end of life, one for the PCB and one for the aluminum cap, that do not exist in the current linear model.

3.1.4 Product Description and Bill of Materials

The competitive benchmark product selected for this study is a commercially available Philips LED A60 E27 bulb. The product was physically disassembled and analysed to establish a detailed bill of materials (BOM), including component masses, material composition, manufacturing processes, supplier assumptions and transport routes. The complete BOM of the incumbent product is provided in Appendix S3 and serves as the reference case for all subsequent environmental and circularity assessments. The final assembly has a total mass of approximately 30 g and consists of seven main subsystems: the optical enclosure, thermal management structure, LED light engine, driver electronics, E27 cap, internal wiring and packaging. The largest contributors to mass and environmental impact are the plastic enclosure, the aluminium heat spreader, the LED board and the driver PCB.

Based on this reference architecture, the LumiNova concept was developed as a modular circular alternative. A second BOM was created to represent the redesigned product and is also provided in Appendix S4. The overall product architecture remains functionally equivalent to the Philips bulb in terms of lighting performance, power consumption (8 W) and luminous output (800 lm), while introducing several design modifications aimed at improving circularity. These include snap-fit assembly instead of permanent adhesive bonding, replaceable electronic modules, increased use of recycled aluminium and copper, bio-based PCB materials, and simplified end-of-life disassembly.

Both BOMs were structured using a hierarchical multi-level approach, allowing each assembly to be decomposed into subassemblies, components and material fractions. This structure enables direct integration into the Life Cycle Assessment (LCA), Material Flow Analysis (MFA), circularity assessment and cost analysis performed throughout this study. In addition, a CAD model of the LumiNova concept and a simplified FEA validation of the modular architecture were developed to support the proposed design and demonstrate its technical feasibility.

3.1.5 CAD Model and Structural Validation

A CAD model of the LumiNova concept was developed to support the proposed modular architecture and provide a geometric basis for the BOM mass estimates. The model comprises four main components: the optical diffuser, the LED board, the driver PCB and the structural support. The clip-based locking system enabling tool-free disassembly is integrated directly into the diffuser and support geometries, as illustrated in Figure 8.

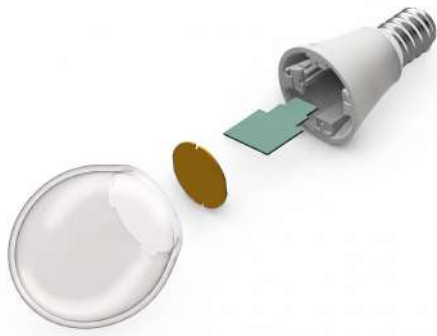


Figure 8: CAD model of the LumiNova modular LED bulb showing the four main components (optical diffuser, LED board, driver PCB and structural support) and the integrated clip-based assembly system enabling tool-free disassembly and component replacement.

To verify the structural integrity of the clip system, a finite element analysis (FEA) was conducted on the optical diffuser clips, which represent the primary mechanical interface during assembly and disassembly cycles. Two load cases were evaluated: the deflection of the clip under insertion force, and the maximum stress at the clip root. Both results confirm that the design remains within acceptable limits for the selected polymer, validating the structural feasibility of the tool-free modular interface.

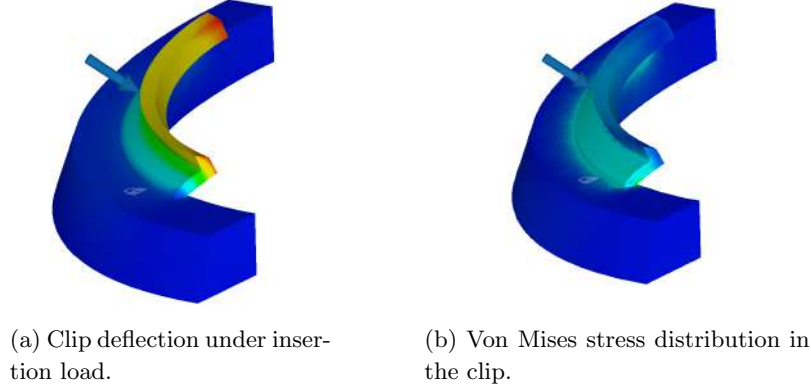


Figure 9: FEA results for the optical diffuser clip system: deflection (left) and stress distribution (right) under insertion loading, confirming that both displacement and maximum stress remain within the design limits of the selected polymer.

3.2 Best Carbon Footprint Materials

3.2.1 Upstream Scope 3 Material Hotspots

The environmental footprint of LED bulbs is strongly influenced by upstream Scope 3 emissions associated with raw material extraction, refining processes, and component manufacturing. A material hotspot analysis was therefore conducted to identify the main contributors to the embodied carbon footprint of the incumbent Philips LED bulb.

The driver PCB and MCPCB (metal-core printed circuit board) represent significant contributors due to the production of epoxy resins, copper electrolysis, and primary aluminium smelting. The LED package itself contains several high-impact materials, including gallium nitride (GaN), phosphor rare earth elements, and gold-based surface finishes. Gold electroplating used in ENIG (Electroless Nickel Immersion Gold) finishes exhibits one of the highest carbon intensities, reaching approximately 40 kgCO₂e/kg due to mining, cyanide refining, and electrochemical deposition processes (Yang & An, 2026).

Gallium nitride is another major hotspot, with reported carbon intensities ranging from 35 to 80 kgCO₂e/kg because of energy-intensive MOCVD epitaxy processes and associated mining activities. Primary aluminium used for the E27 shell and MCPCB substrate contributes approximately 8.5 kgCO₂e/kg owing to the energy-intensive Bayer and Hall-Héroult processes (European Aluminium, 2024). Finally, silicone encapsulants and phosphor materials contribute through petrochemical synthesis and rare-earth extraction activities (Cenci et al., 2020).

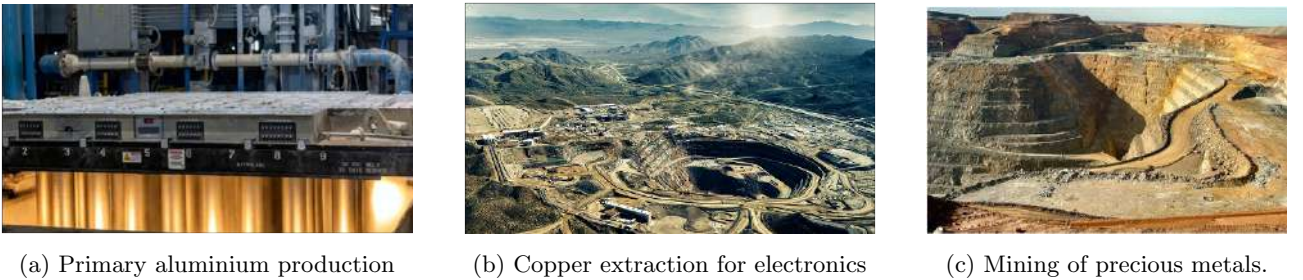


Figure 10: Examples of upstream Scope 3 processes contributing to the embodied carbon footprint of LED bulbs. Aluminium production, copper extraction and mining activities associated with electronic materials represent major environmental hotspots.

3.2.2 Selection of Low-Carbon Materials

The hotspot analysis indicates that the dominant upstream Scope 3 contributors are the LED package (GaN and phosphors), gold surface finishes, the driver PCB, and primary aluminium components. To reduce their emissions, several alternative materials were selected for the LumiNova circular design.

First, primary aluminium used in the bulb shell and MCPCB substrate was replaced by recycled aluminium. Recycled aluminium requires only approximately 5% of the energy needed for primary aluminium production and reduces the carbon footprint from approximately 8.5 to 2.6 kgCO₂e/kg, corresponding to a reduction of nearly 69% (*European Aluminium, 2024*).

Second, the conventional FR4 epoxy resin used in the driver PCB was replaced by an emerging biodegradable resin system. Recent studies have demonstrated the feasibility of bio-based PCB substrates while significantly reducing embodied carbon emissions and facilitating end-of-life treatment. The estimated carbon footprint decreases from approximately 15 to 4 kgCO₂e/kg, representing a reduction of about 67%.

Third, the ENIG surface finish was redesigned to minimize gold content by replacing part of the gold coating with silver and copper-based alternatives. Gold refining is one of the most carbon-intensive processes within the LED value chain; therefore, reducing its use lowers the carbon footprint of PCB surface finishing from approximately 40 to 16 kgCO₂e/kg, corresponding to a reduction of around 40% (*Yang & An, 2026*).

Finally, the modular architecture extends the functional lifetime of the bulb from 15,000 h to 60,000 h. Although this modification does not directly change the material composition, it significantly reduces the amount of material required per functional unit by avoiding complete bulb replacement when a single component fails. This lifetime extension decreases the embodied carbon footprint from approximately 0.327 to 0.196 kgCO₂e per bulb over the considered functional unit.

Overall, the selected material substitutions and circular design strategies substantially reduce upstream Scope 3 emissions while maintaining equivalent lighting performance and supporting the objectives of repairability, modularity, and resource efficiency.

3.2.3 EU end-of-life and waste legislation

As stated before, the EU end-of-life and waste legislation for electronic products is the WEEE. Electronic equipments are divided in 6 categories where one of them is for lamps (*European Commission, 2026b*). Member States must meet a collection target of 65% of the average weight of EEE placed on the market in the previous 3 years, or 85% of WEEE generated (*Environmental Protection Agency Ireland, 2025*).

Moreover, the Waste Framework Directive (WFD) sets the hierarchy for the end-of-life of the light bulb with first prevention, then preparing for re-use, recycling, other recovery, and finally disposal (*European Commission, 2026c*).

In addition, the plastic parts in the product follows the new Packaging and Packaging Waste Regulation (PPWR) that applies from mid-2026 (*European Commission, 2026a*). It says that packaging on EU market must be recyclable by 2030.

Another important piece of EU legislation is the Restriction of Hazardous Substances Directive (RoHS), which complements the WEEE framework by limiting the use of hazardous substances in electrical and electronic equipment (*European Commission, 2025*). Its objective is to reduce risks to both human health and the environment, especially during production, recycling, and waste treatment. This is particularly relevant for LED bulbs, which contain electronic components, solders, polymers, and flame-retardant materials.

There are several possibilities to meet circularity and recycling targets at end-of-life with re-use of recovered materials. One of them is to implement closed-loop recycling system for PCB, aluminium, or copper. Indeed, it would be possible to dismantle those parts from the bulb and the re-use of them as secondary raw materials instead of virgin ones. Furthermore, the re-use of modular parts is another way to meet circularity. The objective is to dismantle the different parts of the product and change only the one that does not work (most of the time the LED or the PCB) and keep the other parts to re-use them for new light bulbs.

3.3 Circular Supply Chain

3.3.1 Manufacturing Route and Circular Supply Chain

LumiNova's manufacturing strategy is built around a clear principle: in-house production is focused on the components that required a redesign for modularity, while standard components that did not need modification are sourced directly from European suppliers. This choice was made to keep the supply chain geographically close to LumiNova's final assembly site in Switzerland, reducing transport-related emissions and improving supply chain resilience, in contrast with major industry players such as Signify whose product-related suppliers are mainly located in Greater China (*Signify, 2025a*). As most suppliers are located in nearby European countries, the majority of component shipments to LumiNova's Swiss assembly site are made by rail, further reducing the carbon footprint of the manufacturing process. The manufacturing route (see Figure 11) is organized around four main modules, each made at a different location in Europe before being sent to *LumiNova* (CH) for final assembly. The PCB driver sandwich is mostly made in-house at *LumiNova* (CH): the biodegradable resin substrate is combined with glass fiber from *3B Fibreglass* (*3B Fibreglass, 2025*) (BE) and recycled metals from *Aurubis* (*Aurubis AG, 2026*) (DE), and pressed using a PCB lamination press bought from *Bürkle* (*Bürkle GmbH, 2026*) (DE) and operated in-house. However, the reflow soldering step requires technical skills that *LumiNova* does not yet have in-house; the gold surface finish components from *Atotech* (*Atotech, 2026*) (DE) are therefore sent directly to *AT&S* (*AT&S, 2026b*) (AT) for soldering, along with the electrolytic capacitor from *Frolyt* (*Frolyt, 2026*) (DE), before the finished assembly is shipped back to *LumiNova* (CH). The LED light engine is also assembled by *AT&S* (*AT&S, 2026b*) (AT) using LED packages from *Ams OSRAM* (*Ams OSRAM, 2026*) (AT), and then sent to *LumiNova* (CH). The optical envelope, including the snap-fit neck system, is made by *Novoplast* (*Novoplast AG, 2025*) (CH) through injection moulding, before being shipped to *LumiNova* (CH). Finally, the E27 aluminium cap is made by *Heggli & Gubler* (*Heggli & Gubler AG, 2025*) (CH) using metal spinning and recycled aluminium from *Novelis* (*Novelis, 2026a*) (CH), and then sent to *LumiNova* (CH). A ceramic insulator from *BJB GmbH & Co. KG* (*BJB GmbH & Co. KG, 2026*) (DE) is also added to the cap assembly. All modules come together at *LumiNova* (CH) for final assembly, followed by quality control and packaging by *DS Smith* (*DS Smith, 2026*) (UK), before being sent to customers through hardware stores and B2B channels.

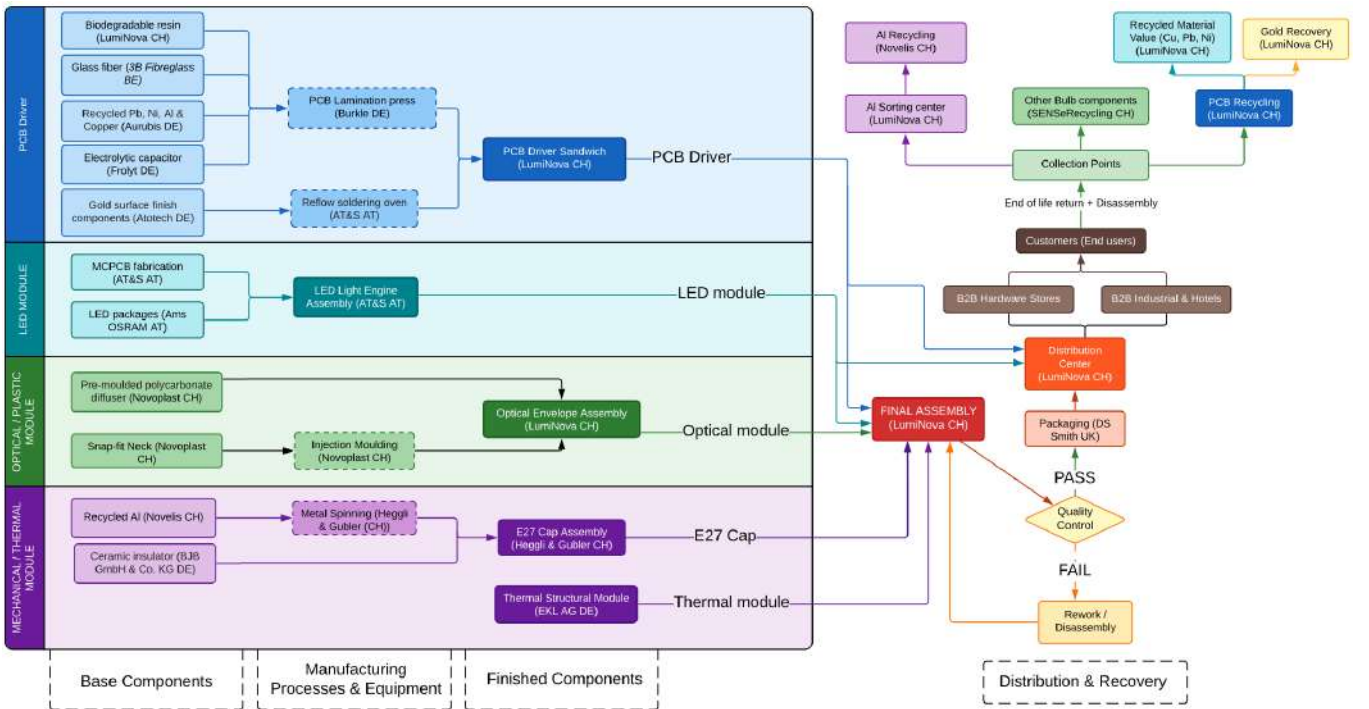


Figure 11: Manufacturing route of LumiNova's circular LED bulb.

Based on this manufacturing route, three supply chain scenarios were developed to illustrate LumiNova's transition from a conventional linear model toward increasingly circular recovery pathways. Starting from a fully

linear system (Figure S5), the scenarios progress to a circular model for 2030 (Figure S6) and finally to a more advanced circular system for 2050 (Figure S7). Together, they demonstrate how improvements in product design and recovery strategies can reduce environmental impacts across the supply chain while enhancing LumiNova’s overall sustainability performance.

The first scenario represents the linear supply chain shown in Figure S5. Since detailed supply chain data from large lighting manufacturers such as Signify is not publicly available, LumiNova’s own product was used as the baseline. In this model, the bulb follows a conventional industrial pathway. Raw materials are extracted, transformed into electronic and structural components, assembled into the final bulb, distributed to the customer, used until failure, and then discarded. At end-of-life, the product is sent to SENSeRecycling, where it is shredded and treated as one mixed waste stream.

As illustrated in Figure S5, the different components of the bulb, such as the PCB, the aluminium cap, and the LED module, are not separated before recycling. This means that valuable materials become mixed together during treatment, making selective recovery difficult. Even if some materials are technically recycled, much of their original value is lost because they cannot be reused at the same quality level. This creates a system where materials quickly leave the product cycle and must constantly be replaced by newly extracted resources.

This creates high impacts across all supply chain scopes. The upstream scope is important because new raw materials must constantly be extracted. The production scope remains high because the complete bulb must be manufactured again even if only one component has failed. The downstream scope is also inefficient because valuable materials are lost during shredding.

The impact of this system on the SDGs remains limited. Because the product follows a simple disposal pathway, there is little contribution to better resource management, industrial innovation, or waste reduction. The linear model therefore has weak positive impact on sustainability and mainly maintains environmental pressure through continuous material consumption.

The second scenario introduces LumiNova’s circular supply chain for 2030, illustrated in Figure S6. This scenario is based on the modular redesign of the product, especially the snap-fit neck system that allows the bulb to be easily disassembled by the user. Instead of discarding the whole product after a failure, customers can identify the failed component and return only the defective PCB, or return the full bulb if necessary, to collection points located in hardware stores. The returned products are then transported to LumiNova’s distribution centre, where they are sorted into dedicated recovery streams. As shown in Figure S6, the aluminium cap is separated and sent to Novelis for recycling, while the PCB is recovered directly by LumiNova to extract copper and other useful materials. The remaining bulb body is still processed by SENSeRecycling.

This modular organisation changes the logic of the supply chain. The product is no longer treated as one disposable object but as a set of recoverable parts, extending product lifetime since only failed components need replacement and creating new value by allowing LumiNova to recover materials directly and reintegrate them into future production. As shown in Figure S6, this redesign reduces impacts across the entire supply chain: the upstream scope decreases because recovered materials partly replace virgin resources, the production scope becomes lower because replacing one failed part requires fewer resources than manufacturing a complete new bulb, and the downstream scope improves because material separation allows better recovery and less waste. The impact on the SDGs becomes clearly more positive, as this system improves resource efficiency, reduces waste generation, and supports innovation through modular design and reverse logistics, creating a stronger contribution to sustainability by demonstrating how circular design can improve both environmental and industrial performance. The third scenario, shown in Figure S7, represents LumiNova’s most advanced circular vision, building on the 2030 system but introducing further technological developments that make circularity more complete, most notably the addition of a fourth recovery stream through in-house LED module recycling.

As illustrated in Figure S7, this becomes possible through the development of a bio-based dielectric layer in collaboration with Sustronics. This innovation simplifies the separation and treatment of the LED module, which is normally one of the most difficult parts of the bulb to recover. In addition, the replacement of gold-based ENIG finishing with Organic Solderability Preservative removes the need for gold and nickel, reducing dependence on critical materials. LED assembly is also progressively brought in-house, giving LumiNova stronger control over both production and recovery.

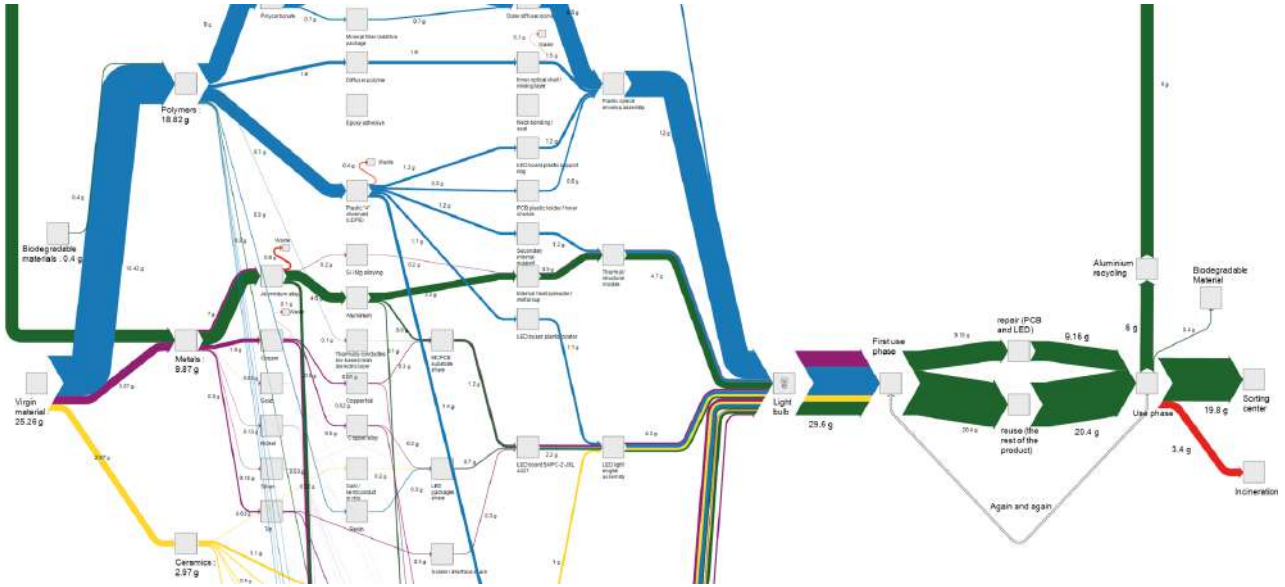


Figure 13: Zoom of the MFA of the new circular product

Several losses are visible on both graphs but they are minor compared to the principal losses that come from the extraction and transformation of the raw materials. Moreover, the losses smaller than 0.1 g were not taken into account.

The evolution of MCI between the linear incumbent (Fig. 14) and the new product (Fig. 15) confirms the gain in circularity of LumiNova. The transition from an MCI of 24.1 % to an MCI of 70.6% can be mostly explained with the increase of lifetime of the light bulb. As explained before, the modularity of the product allows the re-use of most of its parts which make its mean lifetime much higher. An average of 15 000 hours of use phase for the previous bulb was taken into account (*Signify N.V., 2024*). As the structural parts can be used several time, the lifetime of the product increased up to 60 000 hours.

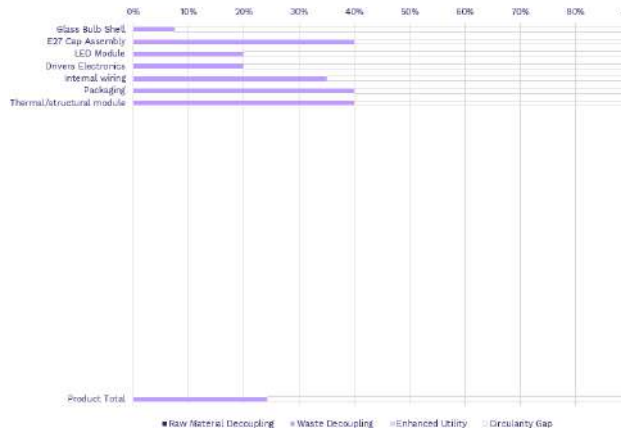


Figure 14: MCI of the linear incumbent

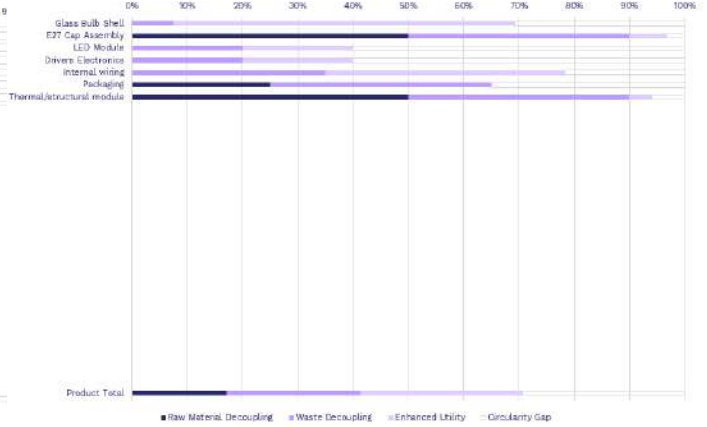


Figure 15: MCI of the new product

3.4 Impact Assessment

3.4.1 Main Environmental, Societal and Human Health Burdens

LED light bulbs generate environmental, societal and human health burdens throughout their life cycle despite their high energy efficiency during operation. To identify the most significant impacts, the incumbent Philips LED bulb was first analysed across multiple environmental and health-related categories. Figure 16 shows that freshwater ecotoxicity, marine ecotoxicity and human carcinogenic toxicity are the dominant impact categories. For all three indicators, the production phase represents the largest contribution, highlighting the importance of material extraction, refining activities and electronic component manufacturing.

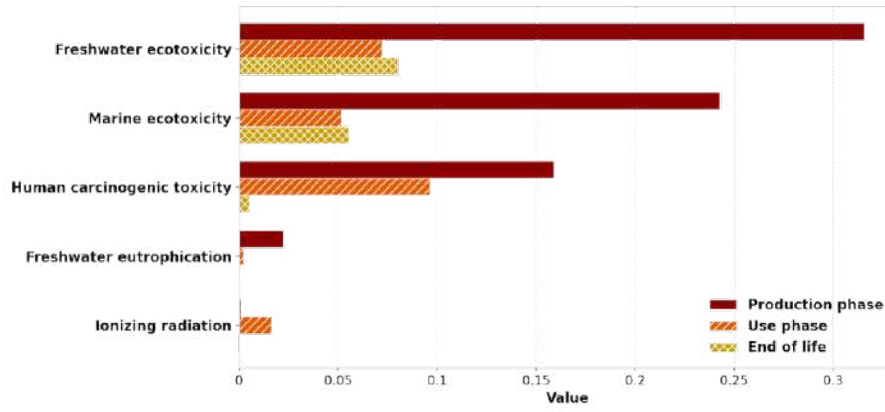


Figure 16: Contribution of the production, use and end-of-life phases to the main environmental and human health impact categories of the incumbent Philips LED bulb.

Freshwater and marine ecotoxicity are mainly associated with the extraction and processing of metals such as aluminium, copper and gold, together with chemicals used during PCB manufacturing. Human carcinogenic toxicity is primarily linked to emissions generated during mining, metal refining and industrial processing operations (Cenci et al., 2020; Illés & Kékesi, 2023; Rahman et al., 2021). These activities also create broader societal concerns including resource depletion, ecosystem degradation, occupational exposure risks and dependence on globalized supply chains.

To evaluate the effectiveness of the proposed circular design, the same impact categories were compared between the incumbent bulb and the LumiNova concept, as shown in Figure 17.

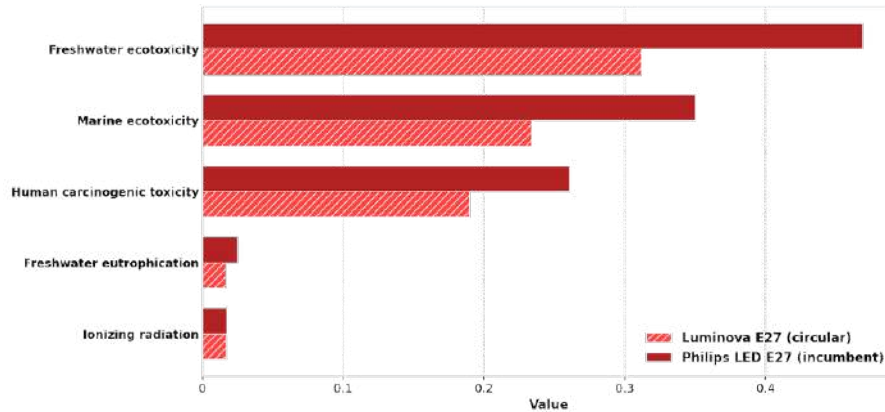


Figure 17: Comparison of the main environmental and human health impact categories between the incumbent Philips bulb and the LumiNova circular alternative.

The results demonstrate a consistent reduction across all impact categories for the LumiNova design. The largest improvements are observed for freshwater ecotoxicity, marine ecotoxicity and human carcinogenic toxicity, indicating that the reduction of virgin material demand and the implementation of circular material strategies directly address the most significant environmental and societal burdens identified for conventional LED bulbs.

Overall, the analysis confirms that the dominant impacts of LED lighting systems originate primarily from upstream material production rather than product use. Consequently, strategies such as recycled materials, improved resource recovery and product lifetime extension represent the most effective levers for reducing environmental burdens while simultaneously addressing societal and human health concerns.

3.4.2 Life Cycle Assessment of the Linear Incumbent and Circular Alternative

The life cycle assessment was performed with SimaPro using the ecoinvent database. The objective was to compare the environmental performance of the linear incumbent LED bulb with the circular LumiNova alternative. The functional unit was defined as:

Providing lighting service for 60,000 hours.

This functional unit was chosen to compare both products on the same lighting service rather than on a single-bulb basis. The system boundaries include raw material extraction, component manufacturing, assembly, transport, electricity consumption during use, and end-of-life treatment.

LCA Modelling Assumptions

The linear incumbent represents a conventional non-disassemblable LED bulb. At end-of-life, the product is treated as a mixed assembly, with limited component separation and material recovery. The circular LumiNova alternative integrates a modular architecture, recycled aluminium and copper, simplified disassembly and dedicated recovery streams for aluminium and PCB materials.

The main modelling assumptions are identical lighting service over 60,000 hours, the same electricity demand during use for both products, replacement of modules instead of complete bulb replacement for LumiNova, the use of recycled aluminium and copper in the circular design, and simplified end-of-life treatment enabled by disassembly.

Climate Change Results

The climate change results are presented in Table 18. The conventional Philips bulb reaches a total impact of **7.21 kg CO₂ eq.**, while LumiNova reaches **5.01 kg CO₂ eq.**. This corresponds to a reduction of **30%**.

$$Reduction = \frac{7.21 - 5.01}{7.21} \times 100 = 30.5\% \quad (1)$$

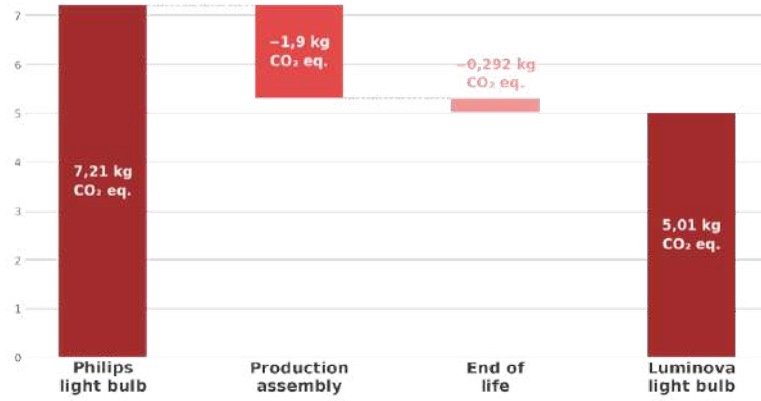


Figure 18: Comparison of total climate change impacts between the linear incumbent and LumiNova.

As shown in Figure 18, the largest reduction occurs during the production phase. The circular architecture avoids unnecessary replacement of functional parts and reduces dependence on virgin materials. This reduction is particularly important because production impacts are directly linked to raw material extraction, electronic manufacturing and global supply chain activities.

Contribution Analysis and Scope Interpretation

The contribution analysis presented in Figure 1 was interpreted using the Scope 1, Scope 2 and Scope 3 framework. Scope 1 emissions are negligible since no direct emissions occur during bulb operation.

Table 1: Interpretation of product-level CO₂e impacts by scope

Scope	Interpretation
Scope 1	Negligible direct emissions during operation.
Scope 2	Electricity consumption during the use phase.
Scope 3 upstream	Raw materials, manufacturing, transport and assembly.
Scope 3 downstream	End-of-life treatment, recycling and waste management.

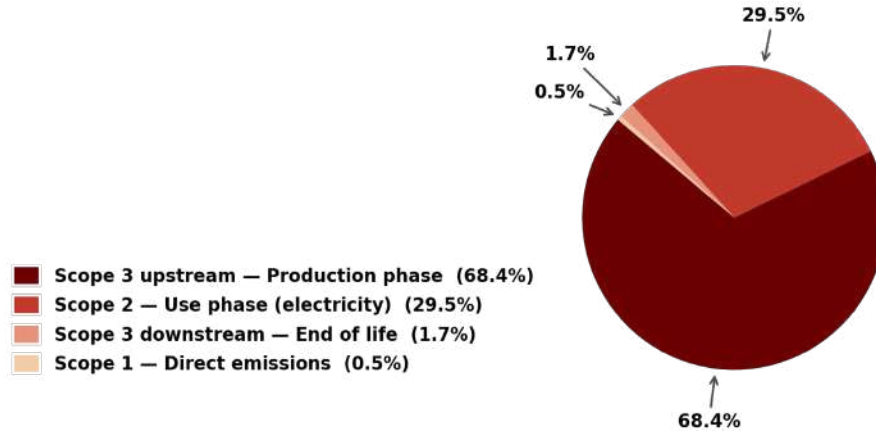


Figure 19: Contribution analysis of the different life cycle stages for both products.

Figure 19 shows that Scope 3 upstream emissions dominate the overall climate change impact, accounting for approximately 68% of total emissions. These impacts are mainly associated with material extraction, electronic component manufacturing and product assembly. Scope 2 emissions represent approximately 30% of total emissions and are linked to electricity consumption during the use phase.

LumiNova reduces production-related impacts by approximately **36%** through recycled materials and modularity. While Scope 3 downstream emissions remain comparatively small, dedicated recovery streams for aluminium and PCB materials improve end-of-life circularity and material recovery.

Monte Carlo and Sensitivity Analysis

A sensitivity analysis was performed to evaluate the robustness of the results and the influence of uncertain parameters. The Monte Carlo results are shown in Figure 20.

The main parameters tested during the sensitivity analysis were the electricity mix during use, recycled aluminium and copper content, electronic component lifetime, end-of-life recycling efficiency, and uncertainty associated with PCB materials such as gold, gallium, terbium and other rare earths.

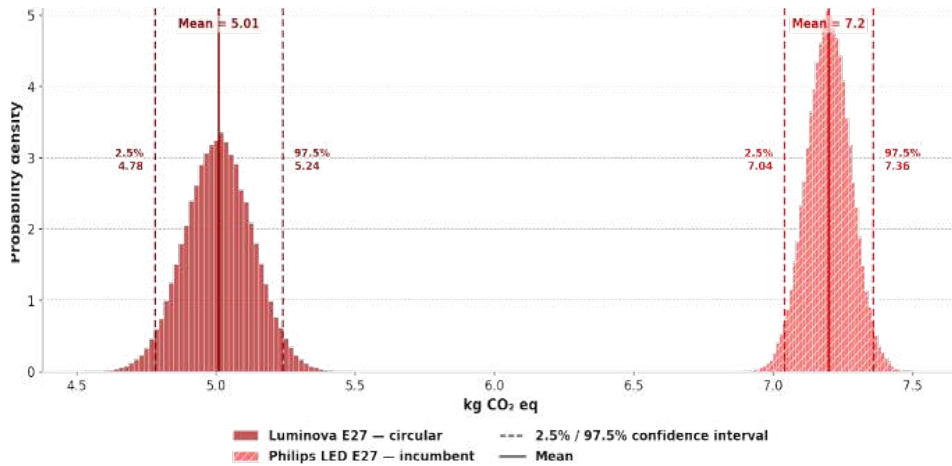
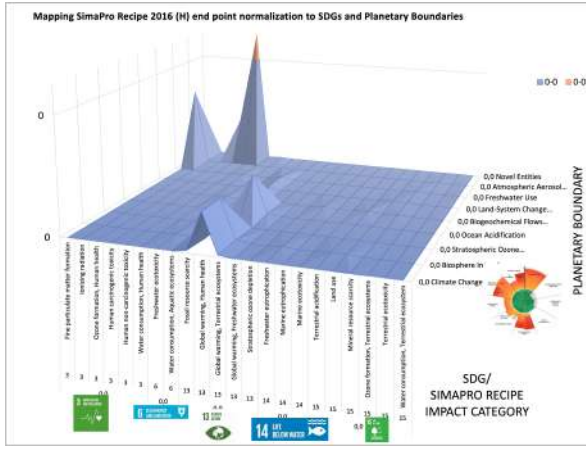


Figure 20: Monte Carlo comparison between the linear incumbent and LumiNova.

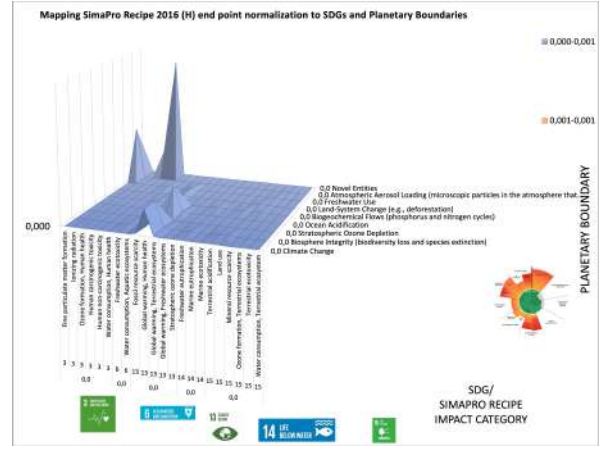
As shown in Figure 20, most simulations favour LumiNova over the incumbent product. This confirms that the reduction is not only linked to one deterministic scenario, but remains robust under varying assumptions.

The electricity mix remains the most sensitive parameter because the use phase dominates the total impact. However, even under different electricity assumptions, LumiNova maintains a lower environmental footprint due to reduced production and end-of-life impacts.

The lifetime of electronic components is also a critical factor. In the linear model, early PCB failure leads to complete bulb replacement. In contrast, LumiNova only requires replacement of the defective module. This makes the circular architecture increasingly advantageous over long operating durations.



(a) Incumbent Philips LED bulb.



(b) LumiNova circular LED bulb.

Figure 23: Mapping of ReCiPe 2016 (H) endpoint normalized impacts to Planetary Boundaries and associated Sustainable Development Goals (SDGs) for the incumbent Philips LED bulb and the proposed LumiNova circular design.

Compared with the incumbent design, LumiNova achieves a systematic reduction across all planetary boundary categories. The largest improvements are observed for Novel Entities (-32.1%), Biogeochemical Flows (-32.3%), Ocean Acidification (-30.6%), Climate Change (-30.5%) and Biosphere Integrity (-30.4%). Atmospheric Aerosol Loading is reduced by approximately 30.0%, while Land-System Change (-25.6%) and Freshwater Use (-25.1%) also show significant improvements. The only category showing a smaller reduction is Stratospheric Ozone Depletion (-15.9%), although its absolute contribution remains negligible compared with the dominant impact categories.

From an SDG perspective, these reductions directly support SDG 13 by lowering greenhouse gas emissions, SDG 14 and SDG 15 through reduced ecosystem pressures, and SDG 3 by decreasing impacts related to particulate matter formation, toxicity, and resource extraction. The reduction in Novel Entities is particularly relevant because it reflects lower consumption of electronic materials and reduced generation of waste streams associated with the production and disposal of LED bulbs.

Overall, the mapping demonstrates that the circular design strategy not only reduces environmental burdens identified through the LCA but also contributes positively to maintaining planetary boundaries within safer operating limits while supporting several sustainability goals related to human health, ecosystems, and resource conservation.

3.5 Cost Model of Product

3.5.1 Manufacturing Cost as a Function of Production Volume

A technical cost model was built directly from the bill of materials (BOM) and the three-cell manufacturing process (Stamp → Inject → Trim/Assembly) in Excel. Three scenarios bracket the likely commercialisation trajectory of the LumiNova modular LED bulb: (a) start-up at **10 000 units/yr** (pilot workshop, pre-series), (b) medium volume at **100 000 units/yr** (first industrial plant), and (c) full scale at **1 000 000 units/yr** (mature national plant). Production duration is 7 years at up to 3 shifts/day, consistent with the physical lifetime of the equipment at that utilisation (Wakeman, 2026). All figures are in euros per part and are based on European industrial benchmarks (Bijleveld et al., 2024; Bruderer AG, 2024; Mekalite, 2025; Montrose Molders, 2024).

Table 2: Manufacturing cost breakdown per bulb by cost driver and production scenario (EUR/part).

Cost driver	(a) 10 k u/yr	(b) 100 k u/yr	(c) 1 M u/yr
Material 1 - recycled Al cap E27 (1.4 g)	0.004	0.004	0.004
Material 2 - polymers + LED electronics (25 g)	2.71	2.39	2.09
Machine depreciation (CAPEX amortisation)	5.79	1.82	0.64
Tooling (moulds + dies)	1.11	0.16	0.11
Plant operation (140 EUR/m ² /yr)	2.66	0.42	0.09
Energy (0.18 EUR/kWh)	0.12	0.12	0.09
Labour - direct + indirect	3.65	2.06	1.26
Consumables	0.04	0.02	0.01
Total manufacturing cost	16.08	7.00	4.30

Three effects govern the strong scale dependence visible in Table 2 and Figure 24. First, **machine depreciation and tooling** dominate at start-up (>7 EUR/bulb combined) because the equipment is heavily under-utilised (machine utilisation $\approx 10\%$ at 10 k u/yr) and the CAPEX is spread over very few parts; both drop by roughly one order of magnitude at full scale. Second, **labour** is the largest variable cost at low volume (manual assembly cell), and remains significant even at full scale because three shifts must be staffed in parallel on the automated line. Third, the **material floor** (≈ 2.1 – 2.7 EUR/bulb) is set by the LED-engine chips, driver IC and polycarbonate optic; it cannot be reduced by volume alone, making supply-chain sourcing agreements critical at full scale.

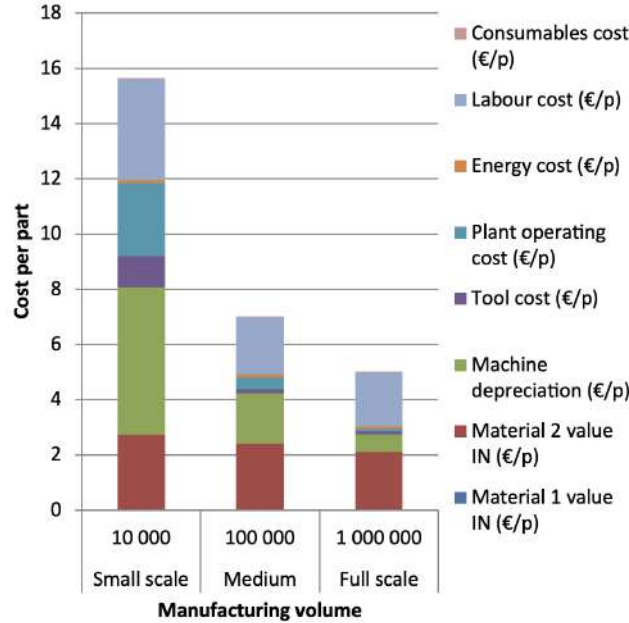


Figure 24: Manufacturing cost as a function of annual production volume, broken down by cost driver.

3.5.2 Gross Margin and Pricing Strategy

A retail price of **9.99 EUR/bulb** was selected to remain competitive with conventional A60 LED bulbs while reflecting the circular value proposition of the modular architecture (Table 3). This yields a gross margin that evolves from a **loss at start-up** (-61%) to **30%** at medium volume and **57%** at full scale. The medium-volume break-even on manufacturing cost alone is reached at approximately **75 000 units/yr**. The modular architecture also opens a second revenue stream: a replacement LED engine or driver PCB is sold at 1.99 EUR, generating additional margin on refurbishment without a new manufacturing cycle.

Table 3: Gross margin at the proposed retail price of 9.99 EUR/bulb.

	(a) 10 k u/yr	(b) 100 k u/yr	(c) 1 M u/yr
Manufacturing cost (EUR/p)	16.08	7.00	4.30
Retail price (EUR/p)	9.99	9.99	9.99
Gross margin	−61%	+30%	+57%

3.5.3 Environmental Shadow Cost

The environmental shadow cost monetises the external damages generated by the circular LED bulb over its full life cycle, converting ReCiPe 2016 endpoint results into euros using the central values of the CE Delft Environmental Prices Handbook 2024 (Bijleveld et al., 2024). Shadow prices represent constructed monetary values assigned to impacts that have no explicit market price, reflecting the loss of economic welfare caused by pollution (Bijleveld et al., 2024). The endpoint monetary factors applied are: 85 000 €/DALY for human health (Bijleveld et al., 2024), 30.1 M€/species·yr for terrestrial ecosystems, and 22.0 M€/species·yr for freshwater and marine ecosystems (Bijleveld et al., 2024). Resource scarcity scores (in USD₂₀₁₃) are converted at approximate parity.

The monetised results are summarised in Table 4. The total shadow cost of the bulb is **4.03 €/unit**, distributed across three main domains. Human health accounts for 88.5% (3.62 €), driven predominantly by human non-carcinogenic toxicity (1.85 €) and fine particulate matter formation (0.80 €), both attributable primarily to the manufacturing (assembly) phase as identified in the process-level LCA results. Resource scarcity contributes 10.2% (0.42 €), with fossil resource depletion (0.34 €) reflecting the energy-intensive production chain. Ecosystem impacts remain negligible (<0.01 €), as the impacted species·yr quantities are infinitesimally small despite the high monetary value per unit.

Table 4: Environmental shadow cost by impact domain

Domain	Shadow cost (€/unit)	Share (%)
Human health (DALY)	3.62	88.5
<i>of which non-carc. toxicity</i>	<i>1.85</i>	<i>45.9</i>
<i>of which fine particulates</i>	<i>0.80</i>	<i>19.8</i>
<i>of which global warming</i>	<i>0.40</i>	<i>9.8</i>
Terrestrial ecosystems (species·yr)	0.001	0.03
Freshwater & marine ecosystems	0.000	0.01
Resource scarcity (USD ₂₀₁₃)	0.42	10.2
Total	4.03	100

These results highlight that the dominant environmental liability of the LED bulb lies in its manufacturing phase, where material processing generates significant toxic emissions. A circular redesign targeting component reuse or substitution of hazardous substances in the assembly stage would therefore yield the greatest reduction in shadow cost.

3.5.4 CAPEX by Production Scale

CAPEX was detailed cell by cell for the three scenarios (Table 5, Figure 25). The three in-house production cells are: **Stamp** (progressive die stamping of the recycled aluminium E27 cap), **Inject** (polycarbonate optical envelope and supports, multi-cavity mould), and **Trim/Assembly** (modular snap-fit assembly, ICT electrical test, packaging).

Table 5: CAPEX detailed by production cell for the three scenarios (EUR). Equipment costs are sourced from published market data (*Bruderer AG, 2024; IMARC Group, 2025; Mekalite, 2025; Montrose Molders, 2024*); tooling costs from injection-mould pricing guides (*Mekalite, 2025; RapidDirect, 2026*).

Production cell	(a) Start-up	(b) Medium	(c) Full scale
Stamp - Al cap E27 (press + tooling)	80 000	150 000	400 000
of which die/tooling	20 000	25 000	40 000
Inject - PC envelope (press + mould)	150 000	350 000	1 400 000
of which mould	50 000	70 000	240 000
Trim - Assembly + test + packaging	80 000	350 000	1 200 000
Total equipment CAPEX	310 000	850 000	3 000 000
Total project	388 000	960 000	3 450 000

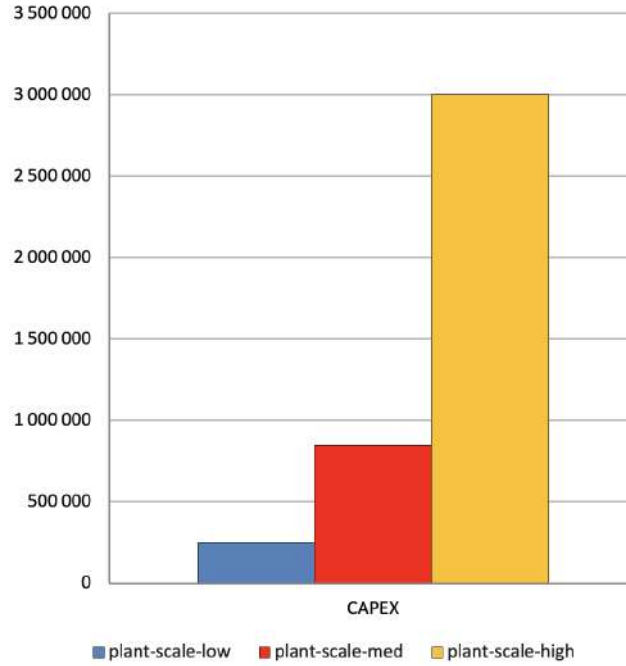


Figure 25: CAPEX by production scale for the three scenarios.

The CAPEX scales super-linearly with volume because the Inject cell saturates a single press at ≈ 100 k u/yr, requiring a second parallel press at full scale. Equipment is depreciated linearly over 7 years (3-shift plant life (*Wakeman, 2026*)); injection moulds are depreciated over their tool life (1–2.5 million shots depending on steel grade).

3.5.5 Stakeholder Engagement and NetPositive Investments

To become a NetPositive company, LumiNova will reinvest part of its future profits into three strategic stakeholders that directly influence the environmental and societal performance of its circular supply chain.

1. Closed-Loop Aluminium Partnership (Novelis) Novelis is the supplier of recycled aluminium used in the E27 cap assembly and is already integrated into LumiNova’s manufacturing route (*Novelis, 2026b*). To strengthen circularity, LumiNova will invest **€100,000 in 2027** to establish a dedicated closed-loop recycling programme and improve traceability between recovered aluminium caps and new bulb production. An additional **€50,000 per year from 2028 to 2030** will be allocated to collection logistics, sorting infrastructure, and certification of secondary aluminium streams.

The objective is to increase recycled aluminium content from 70% to more than 95 by 2030 while reducing dependence on virgin aluminium production. This investment directly supports SDG 12 (Responsible Consumption and Production) and SDG 13 (Climate Action).

2. Sustainable PCB Development Partnership (AT&S) The PCB driver is one of the main environmental hotspots identified in the Life Cycle Assessment. LumiNova therefore plans a strategic partnership with **AT&S**, the Austrian electronics manufacturer responsible for the assembly of both the PCB driver and LED module (AT&S, 2026a).

A total investment of **€250,000 between 2027 and 2029** will support joint R&D activities focused on biodegradable PCB substrates, gold-free Organic Solderability Preservative (OSP) finishes, and improved module disassembly. The funding will be used for pilot-scale manufacturing trials, reliability testing, and industrial qualification.

The stakeholder aims to reduce embodied carbon emissions, decrease dependence on critical raw materials, and improve end-of-life material recovery.

3. Retailers and End-Users Take-Back Network LumiNova’s circular business model relies on the return of defective modules through partner hardware stores and distribution channels. To support this ecosystem, LumiNova will invest **€150,000 in 2027** to deploy collection points at retail partners and establish a digital reverse-logistics tracking platform. In addition, **€20,000 per year from 2028 onwards** will be dedicated to retailer training, consumer awareness campaigns, and financial incentives for returned modules.

This programme directly involves hardware stores, industrial customers, and end-users, transforming them into active participants in the circular supply chain. Higher return rates will increase component reuse, material recovery, and product lifetime extension.

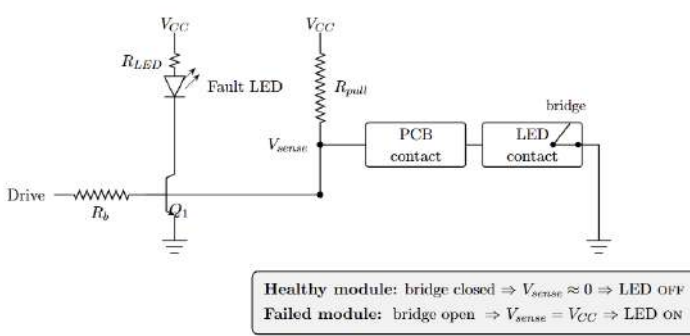
3.6 Initiatives

3.6.1 SMART Initiatives

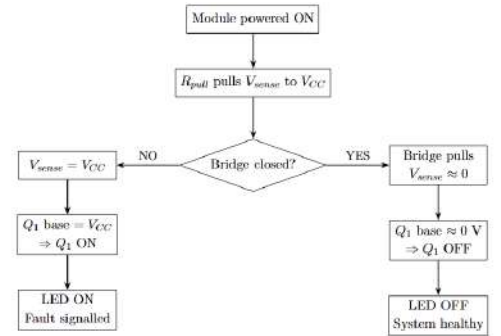
The modular design is the conceptual and technical basis of the project, but it will require several prototype cycles, expert input, and capital investment before it can be validated. Once established, recycled aluminum can be introduced to reduce the product’s carbon footprint. In parallel, PCB resin substitution and gold reduction should be screened over the next 1–2 years, with 2030 as the target horizon. The most demanding objectives, including full gold elimination and a localized supply chain, are reserved for the long term and will depend on further research and industrial scaling. The brainstorming process and a summary of the resulting initiatives are presented in the Appendix (Figures S10 & S11).

Initiative 1: Fully modular LED bulb design

The first initiative focuses on a fully modular LED bulb architecture. Since the dominant failure modes of LED lighting are chip heating and material aging, the objective is to enable component-level replacement of the PCB driver and LED board while eliminating permanent epoxy-based sealing (Luger Research, 2010; Luiten, 2013; Mitterhuber et al., 2020). To support repairability, the design includes built-in fault detection using a shunt resistor and photodiode to identify whether the fault lies in the PCB driver or the LED board (Figure 26). The diagnostic logic follows the methodology proposed in (Gupta & Roy, 2020), but is simplified to focus only on the two replaceable modules, keeping the system affordable and suitable for consumer-level repair.



(a) Built-in fault detection architecture.



(b) Decision logic of the bridge-based fault detection system based on (Gupta & Roy, 2020).

Figure 26: Integrated fault detection system enabling the modular repair strategy of the LumiNova LED bulb. (a) System architecture, (b) fault detection algorithm. A closed bridge indicates that all modules are functional, whereas an open bridge activates the fault indicator LED.

Initiative 2: Reduction of the environmental footprint of the PCB

Printed circuit boards contribute significantly to the bulb’s environmental impact, mainly because of gold content, epoxy-based resin, and the difficulty of recovering embedded metals. The second initiative therefore aims to reduce gold use, identify alternative surface finishes, substitute the conventional epoxy substrate with a biodegradable material, and improve recyclability. Initial efforts should focus on gold reduction and biodegradable resin, with full gold elimination as a long-term objective. Current academic research provides a strong proof of concept for biodegradable PCB substrates. In particular, Geiger *et al.* at Empa developed a wood-based substrate that is fully biodegradable and has been demonstrated in a functional device prototype, as illustrated in Figure 27) (Dudnyk *et al.*, 2025).

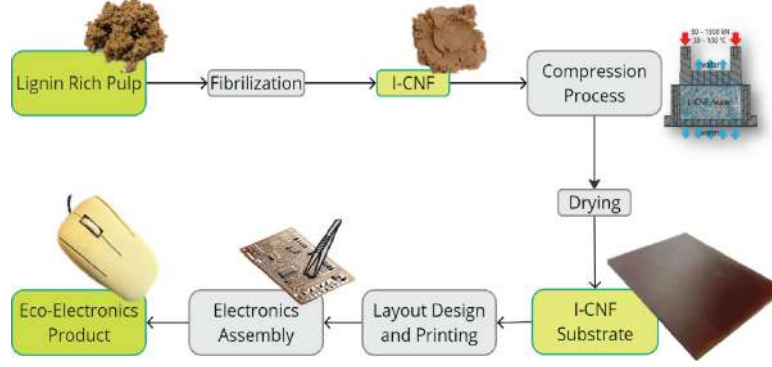


Figure 27: Schematic overview of lignin-rich pulp fibrillation into LCNF and subsequent processing into a rigid, fully biodegradable PCB substrate demonstrated in a computer mouse prototype (Dudnyk *et al.*, 2025)

A practical implementation would require defining the board’s electrical, thermal, and mechanical requirements, screening suitable biodegradable substrates, fabricating prototypes, and testing signal integrity and reliability. Close collaboration with suppliers and manufacturers will also be necessary to ensure process compatibility.

A fully gold-free surface finish remains a long-term objective, targeted for 2050, because current alternatives still face limitations in cost, shelf life, oxidation resistance, solderability, and interconnect reliability (summary of alternatives in Figure 28 (Scharf & Tao, 2012)). Immersion silver is a potential intermediate option, as it is gold-free and lead-free (Long & Toscano, 2013). However, its implementation is constrained by limited shelf-life and susceptibility to oxidation and tarnishing, which can reduce long-term reliability (Long & Toscano, 2013). Ideally, the project would integrate an OSP surface finish as it is free of heavy metals, however, current OSP formulations exhibit poor resistance to handling, humidity exposure, repeated contact, and thermal loading (Rowland & Prasad, 2015). Further materials research and collaboration with academic partners will therefore be required to validate a reliable gold-free solution.

Comparison	HASL	OSP	IAg	ENIG	Hard-Gold	ENIG/SM OBC
Surface co-planarity	Poor	Best	Good	Best	Best	Good
Solder contrast (Assembly)	Poor solder/solder	Best solder/copper	Good solder/silver	Best solder/gold	Good solder/gold	Best solder/gold
Electric contact use	Fair	Poor	Good	Good	Good	Good
PCB high-frequency loss	Fair	Best	Best	Poor	Poor	Good
Environmental Hazard	High (Pb)	Low (Water based)	Medium (Ag)	Medium (Ni)	Medium (Ni)	Medium (Ni)
Cost of surface finish	Standard (1x)	Lowest (0.2~0.3x)	Standard (1x)	Highest (1.5~2x)	Highest (1.5~2x)	High (1.3x)

Figure 28: Comparison of PCB surface finishes (Scharf & Tao, 2012)

Initiative 3: Circular material recovery and recycling

The third initiative focuses on circular material recovery through secondary aluminium and the recovery of valuable metals from end-of-life components. Recycled aluminium is especially suitable for LED housings because it retains the thermal and mechanical properties required for heat dissipation while using substantially less energy than primary aluminum (Luminate Lighting Group, 2025). Printed circuit boards represent a more complex recovery challenge because of their multi-material composition, however, physical pre-treatment

and hydrometallurgical processes have shown high recovery rates for metals such as copper, nickel, and lead (process depicted in Figure 29) (Kumari et al., 2024). These approaches have shown promising laboratory-scale performance and may be suitable for future commercial development. Recovering these metals would also create economic value. In the short to medium term, LumiNova should therefore collaborate with external recycling partners, while internal recovery capabilities could be considered later.

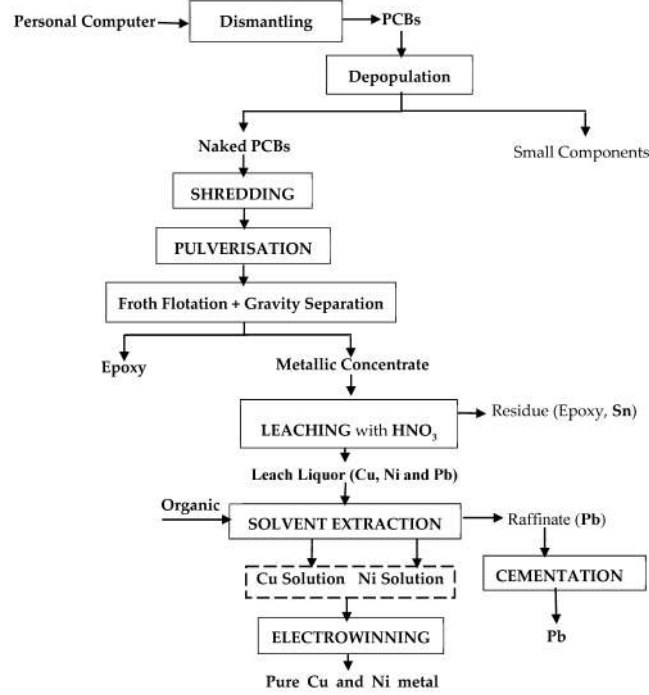


Figure 29: Process flow diagram for metal recovery from printed circuit boards (PCBs). (Kumari et al., 2024)

Gold recovery is particularly relevant because gold contributes disproportionately to the environmental footprint of PCBs while also remaining economically valuable. As a result, reducing gold use and recovering gold from end-of-life components would support both LumiNova's 2030 emissions target and its circularity objectives. Recent research suggests that gold recovery from waste PCBs may become economically viable in 2 years time (Yang & An, 2026). In parallel, Raffaele Mezzenga's work at ETH Zurich on protein amyloid nanofibrils derived from whey side-streams demonstrates a promising bio-based route for high-purity gold recovery from e-waste, as shown in Figure 30 (Peydayesh et al., 2024). These materials have shown high gold adsorption capacity and selectivity, and life-cycle and economic assessments further support the feasibility of this approach (Peydayesh et al., 2024). This local research base is strategically useful for future development and collaboration.

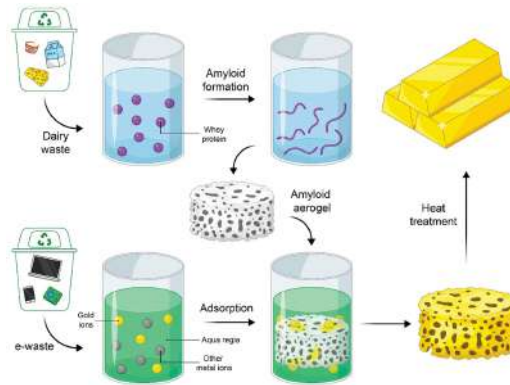


Figure 30: Schematic route of AF aerogels for the selective recovery of gold from e-waste (Peydayesh et al., 2024)

Initiative 4: Local production of PCBs and LED chips

The fourth initiative aims to progressively localize PCB production and, in the longer term, LED chip fabrication. PCB in-house production is targeted for 2030, since several manufacturers already operate in Switzerland

and Europe, making pilot production comparatively feasible (Zhang, 2025). The implementation strategy should begin with supplier qualification and pilot runs before scaling up. LED chip production is more challenging and is therefore treated as a long-term objective, with in-house fabrication targeted for 2050. At present, LED chip sourcing remains highly concentrated in Asia, which limits the feasibility of rapid regional substitution. The extent of supply-chain dependence is illustrated in Figure 31, which shows the EU's reliance on external suppliers for a range of critical raw materials (Cerutti & Nardo, 2023). This dependence reinforces the need for dual sourcing, capability building, and gradual localization. Overall, this initiative is strategically relevant because it would reduce exposure to global supply-chain shocks and is consistent with broader European policy trends, such as the European Chips Act and related efforts to strengthen autonomy in critical technologies (European Parliament & Council of the European Union, 2023).

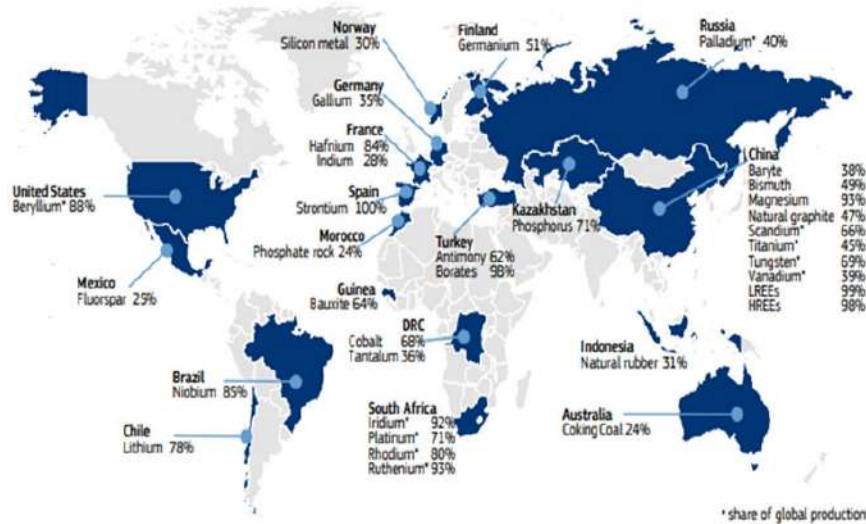


Figure 31: Current EU reliance in relation to raw or processed critical materials (2020 data) needed for Semiconductors (Cerutti & Nardo, 2023)

3.6.2 Change management challenges/barriers and stakeholders to engage

Across the four initiatives, the main change management barriers are technical validation, industrial and supply-chain compatibility, and market adoption. Technical validation is needed to ensure that modularity, biodegradable materials, and metal recovery perform reliably under real-world conditions. Industrial compatibility matters because the solutions must fit existing manufacturing and sourcing systems. Market adoption will determine whether users and institutional buyers are willing to support scaling. A summary of the barriers and stakeholder groups is provided in the Appendix (Figures S12 and S13).

The first barrier concerns technical feasibility. The modular bulb requires iterative prototyping and testing to confirm that robustness is not compromised and that the product can withstand repeated use and component replacement. The biodegradable PCB must show that alternative substrates can meet electrical, thermal, and reliability requirements, while metal recovery must prove efficient and economical beyond the laboratory scale. Key stakeholders include Petr Veselý, whose expertise in soldering, assembly reliability, and sustainable electronics is directly relevant; Géczy Attila, whose research on biodegradable substrates, circular electronics, and WEEE metal recovery provides strong scientific validation and Raffaele Mezzenga, whose work on protein amyloid nanofibrils for gold recovery from e-waste offers a valuable insight into circular material recovery (Peydayesh et al., 2024). These stakeholders can validate whether the proposed solutions are technically credible and where the main performance risks lie.

The second barrier concerns industrial compatibility. Even if the concept works technically, it must fit existing manufacturing, procurement, and supplier systems. The modular bulb needs a production model that supports recycling, repair and replacement, while the biodegradable PCB may require modified fabrication and finishing processes that may not be compatible with standard production lines. Volutio could support LumiNova on recycling, take-back, compliance, and recovery pathway design (Volutio, 2026). In particular, Andreas Bill's

expertise in e-waste recycling and national waste-system development would be valuable for assessing whether the proposed recovery pathway can meet regulatory and operational requirements (*Volutio, 2026*).

Local production also faces supply-chain constraints, especially because PCB and LED manufacturing depend on industrial ecosystems that are not equally available across regions. Figure 32 illustrates how concentrated global chip capacity remains, highlighting why localization is a long-term challenge and why phased implementation and regional partnerships are necessary (*of Auditors., 2025*).

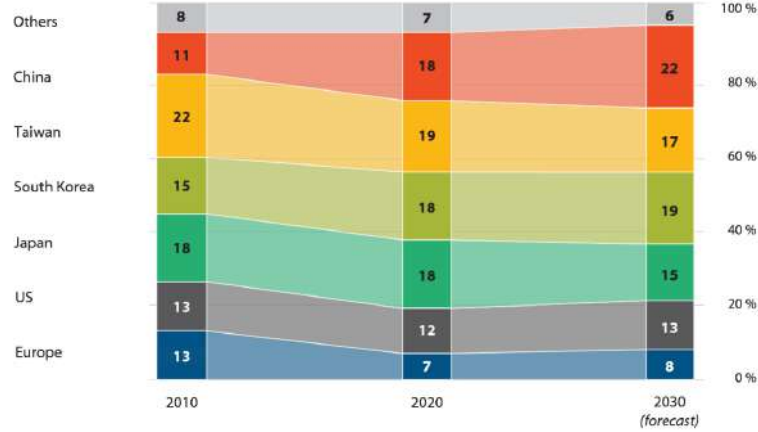


Figure 32: Share of global chip capacity by region in 2010-2030 (*of Auditors., 2025*)

This barrier should be addressed through design-for-manufacturing reviews, supply-chain mapping, and phased collaboration with industrial partners. David Spitz's work on design for manufacturability shows that PCB designs should be optimized early, as the cost of correcting manufacturability problems rises sharply over time. Figure 33 illustrates this principle by showing how the cost of unreliability escalates from concept to production (*Spitz, 2024*). The first step should be to identify local PCB suppliers that can support pilot production and low-volume builds. At the same time, the design should remain compatible with standard manufacturing wherever possible. Local production should start with PCB localization before progressing to LED-chip fabrication.

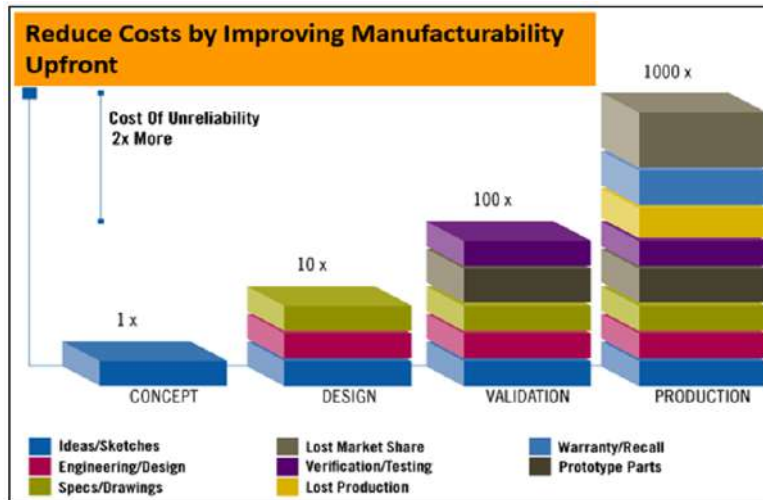


Figure 33: Cost reduction by addressing issues early in product development (*Spitz, 2024*)

The third barrier is acceptance by buyers, users, and organizations. Even a technically strong product may fail if customers perceive it as complicated, unreliable, or inconvenient, especially if modularity is associated with extra maintenance. Institutional buyers are a better entry point because they can evaluate cost, serviceability, and procurement benefits more easily than individual consumers. The adoption strategy should begin with institutional customers, supported by repair guidance and messaging focused on durability, resource efficiency, and reduced costs. Key stakeholders include François Gerber, who leads Circular Economy Switzerland and can help provide visibility and stakeholder connections, Anis Nassar, who leads circular-economy activities at the World Economic Forum and can provide business engagement, and networks such as Lighting for Good, which can help build legitimacy and collaboration within the lighting sector (*Lighting for Good, 2024*). In addition,

the Swiss Federal Office for the Environment (BAFU/FOEN), particularly its Circular Economy sector, could strengthen policy alignment and credibility in relation to Swiss waste-prevention and resource-efficiency goals (*Federal Office for the Environment FOEN, 2026*). These stakeholders are important because they can support trust-building, ecosystem alignment, and the transition from concept to accepted practice.

Securing stakeholder buy-in will require showing value across finance, engineering, operations, and sustainability criteria. For finance and business stakeholders, the case rests on reduced warranty exposure and lower end-of-life liability. For engineering and operations, manufacturability, reliability, and supply resilience must be demonstrated. For sustainability stakeholders, the project must show clear gains in repair, reuse, recycling, and recovery. Finally, LumiNova’s initiatives should demonstrate alignment with emerging industrial-policy shifts.

3.6.3 Absolute Targets

As seen in the roadmap below (Figure 34), LumiNova should achieve a 36% reduction in emissions per bulb by 2030, mainly through gold reduction, increased recycled aluminum, full modular design and implementation of a biodegradable PCB resin. By 2040, emissions should fall by 51% as higher recycled-content materials, and local supplying of PCBs and LED chips are introduced. By 2050, the target should reach a 78% reduction, corresponding to approximately 1.2 kgCO₂e per bulb, enabled by mature circular supply chains, local production, and a complete removal of gold. Table 6, summarises the quantified targets and associated recommendations.

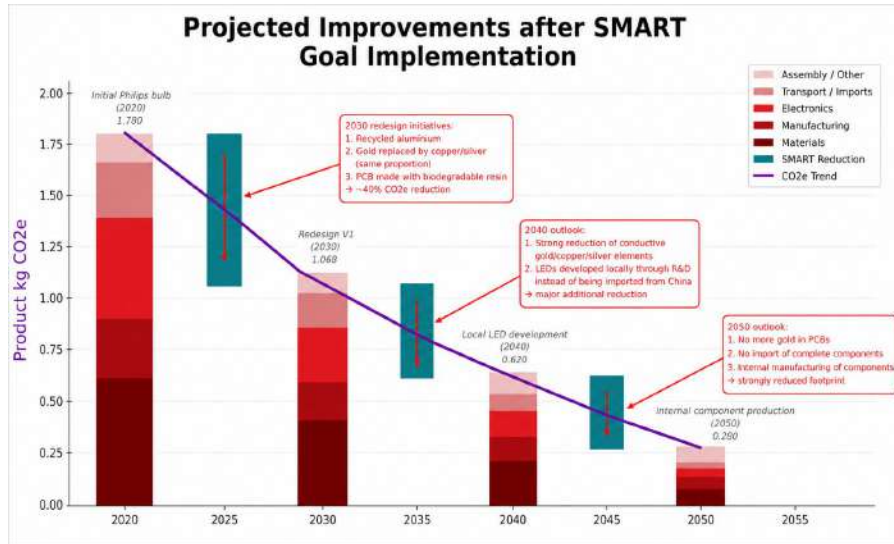


Figure 34: Quantified Absolute Targets

Table 6: Quantified roadmap for emissions reduction and circularity targets

Year	Target	Recommendations
2030	- Reduce gold by 80%. - Recover $\geq 35\%$ of EoL metals. 100% recycled aluminum - Full modular design with $> 2\times$ lifetime. - Biodegradable PCB resin.	- Prioritise gold minimisation in non-critical connections. - Enable repair, replacement, and reuse through modularity. - Begin structured end-of-life metal recovery.
2040	- Recover $\geq 45\%$ of EoL metals. - Achieve $\geq 19\%$ total recycled content. - PCB pilot line of 50 000 parts	- Scale metal recovery from PCBs. - Qualify one local PCB pilot line. - Expand recycled material use. - Strengthen take-back systems.
2050	- Recover $\geq 55\%$ of EoL metals. - Achieve $\geq 35\%$ recycled content. - Full local PCB and LED fabrication.	- Secure local suppliers for in-house fabrication - Close material loops for key metals. - Complete removal of gold in PCB

This roadmap differs from companies such as Dialight because the dominant decarbonisation lever is not use-phase electricity but upstream material and supply-chain redesign. Dialight’s net-zero strategy is shaped by the fact that 89% of its footprint comes from customer use-phase electricity, which makes operational efficiency and grid decarbonisation central to its pathway (*Dialight, 2026*). In LumiNova’s case, the use phase accounts for only 29.5% of total emissions, so the decarbonisation potential is concentrated in material substitution, manufacturing efficiency, and end-of-life recovery. Consequently, LumiNova’s roadmap places greater weight on gold minimisation, recycled metals, PCB redesign, and local component production than Dialight’s product strategy does. Dialight aims to reach net-zero by 2040, with near-term targets to reduce Scope 1 and 2 emissions by 37.8% and Scope 3 emissions by 22.5% by 2029 (*Dialight, 2026*).

Whitecroft Lighting is a relevant circularity benchmark because it targets a 80% reduction in virgin materials by 2030, and is committed to net zero by 2045 with a UK-based supply chain (*Whitecroft lighting, 2026*). Its circular model focuses on refurbishing, repurposing, redistributing, reselling, and recovering luminaires, which is especially relevant given its institutional market focus (*Whitecroft lighting, 2026*). Compared with LumiNova, Whitecroft is stronger on service-based circularity and supply-chain localization, while LumiNova is more ambitious in technical material redesign.

Holophane’s EARTHlight strategy is strongest in packaging reduction, operational circularity, and manufacturing decarbonisation. The company reports a reduction in purchased plastic from 7,000 kg to 627 kg between 2016 and 2025 due mainly to a shift to fully recyclable packaging materials (*Holophane Europe Limited, 2026*). It also powers up to 85% of its on-site energy consumption through onsite solar PV, with the remaining 15% sourced from Carbon Trust-accredited renewable electricity (*Holophane Europe Limited, 2026*). In addition, Holophane designs products for upgradeability, repair, and disassembly, which strengthens their lifecycle-oriented approach. However its claims are less transparent regarding the circular treatment of core electronic components such as LEDs and PCBs, so compared with LumiNova it appears less advanced in upstream component redesign and critical-material transparency.

4 Conclusion

This report has shown that LumiNova, a modular and repairable LED light bulb, offers a practical and scalable circular alternative to the linear use-and-dispose model that still dominates the lighting industry. By redesigning the product around circularity principles, the project reduced the lifecycle carbon footprint from 7.21 kg to 5.01 kg CO₂-eq and increased the circularity index from 24% to 71%.

The main work focused on a modular snap-fit design that addresses the limitations of conventional LED bulbs. This enabled design for disassembly, making it possible to repair or replace individual components rather than discarding the entire product, as well as the extension of the bulb’s service life from 15,000 to 60,000 lumen hours. This was complemented with lower-impact material choices, including a biodegradable resin PCB substrate, a gold-free surface finish, and 100% recycled aluminum, as well as supply-chain strategies such as take-back and replacement systems, dedicated recycling streams, and localised production.

A key finding of the study is that the largest environmental burdens of LED lighting come mainly from upstream material production, particularly metal extraction, refining, and electronics manufacturing, rather than the use phase itself. For this reason, extending product lifetime through modularity proved to be the most effective strategy. The work also showed that this circular model is economically promising, with the financial analysis indicating a strong Net Present Value of 13.53 M€ and a high Social Return on Investment (Tables S2 and S4).

The LumiNova approach can be applied beyond lighting to other consumer electronics facing similar issues of waste generation, repairability, and material recovery. Future development could focus on regional independence by localising PCB and LED chip production, helping to reduce transport emissions and align with the European Chips act. The modular concept also anticipates upcoming EU repairability and removability requirements (Law in force starting February 2027), making it relevant for future regulatory compliance. In addition, the recovery framework developed in this project could be extended to more complex electronic products to support the recovery of critical materials such as copper, nickel, and gallium.

Overall, LumiNova demonstrates that modular design, localised production, and dedicated recovery systems can form the basis of more circular products. By combining technical redesign with material efficiency and supply-chain rethinking, the lighting industry can move closer to meeting environmental legislation while accelerating the transition toward a more sustainable circular economy.

Authors' contribution

The work presented in this report was carried out collaboratively by all team members, with each author taking primary responsibility for specific sections of the project.

Rémy was responsible for the business-oriented analyses, including the Business Model Canvas and the Supply Chain assessment. Léa developed the Supply Chain Map, the Roadmap, the Stakeholder Analysis, and the proposed Initiatives. Elias conducted the E-Sankey analysis and the Circularity assessment. Mathélyo was responsible for the Life Cycle Assessment (LCA) and the development of the Bill of Materials (BOM). Thomas developed the Cost Model and contributed to the Product Design analysis.

All authors contributed to discussions, interpretation of results, and project development throughout the study. All authors reviewed and approved the final written document.

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5 Appendix

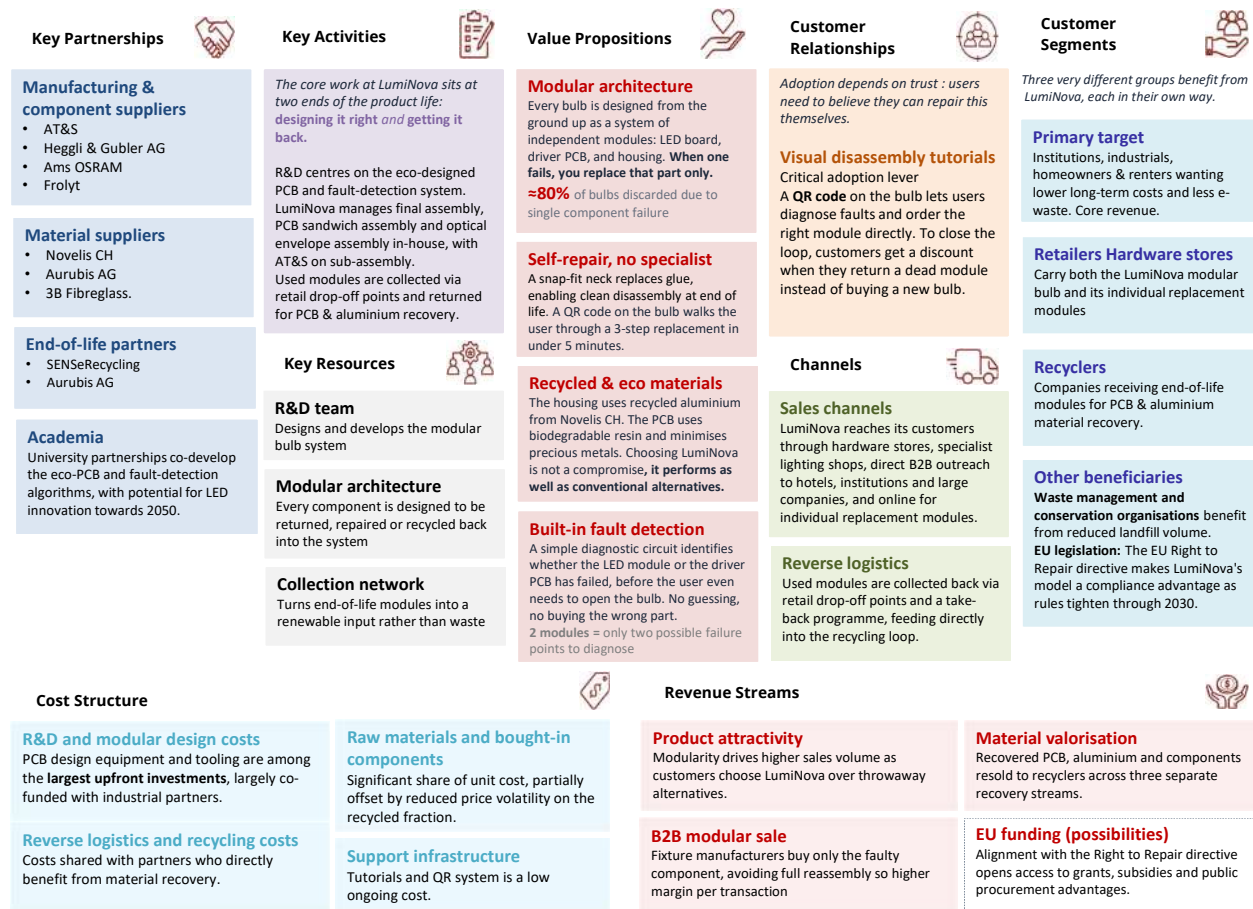


Figure S1: Business Model Canvas



WORKSHEET

Safe & Circular Product Redesign Workshop

Designed for : LumiNova -
Circular LED Bulb

Designed by :
Group H

Date :
14/05/2026

Version : III

EXPLORE (5 MINS)

CHOOSE A COMMERCIAL APPLICATION

Philips LED A60 E27 bulb (SKU 8720169191655)

WHAT ARE THE FUNCTIONAL NEEDS AND REQUIREMENTS OF A LIGHT BULB?

INTERVIEW A USER

Core function of the light bulb:

- Provide efficient, safe, and long-lasting illumination

User needs:

- Reliable lighting
- Low energy consumption
- Long lifetime
- Easy replacement
- Affordable cost

DEFINE THE CHEMICALS OF CONCERN (15 MINS)

PICK & SCREEN TWO CHEMICALS THAT ARE IN THE PRODUCT

This is a summary of all the chemicals that can be found in a product like this, and not a complete list.
Screen the different chemicals using [MaterialWise.org](https://www.materialwise.org).

WHICH CHEMICALS SHOULD BE DESIGNED OUT?

Highlight the chemicals of concern that should be designed out in the next phase.

☐	PCB / ADHESIVE EPOXY RESIN	25068-38-6	TOXIC: (Y) / N
☐	ELECTROLYTE (CAPACITOR) ETHYLENE GLYCOL	107-21-1	TOXIC: (Y) / N
☐	LED SEMICONDUCTOR GALLIUM NITRIDE	7440-50-8	TOXIC: Y / (N)
☐	PHOSPHOR COATING RARE EARTH MIX	N/A	TOXIC: Y / (N)
☐	GOLD (PCB PLATING) ELECTROLESS NICKEL IMMERSION GOLD (ENIG)	7440-57-5	TOXIC: Y / (N)

Figure S2: Safe & Circular Product Redesign Workshop



THE
CIRCULAR
DESIGN
GUIDE



DESIGN TO DESIGN
PRODUCTS
INNOVATION
INSTITUTE

WORKSHEET

Safe & Circular Product Redesign Workshop

REDESIGN (30 MINS)

DEVELOP 3 CONCEPTS FOR IMPROVEMENTS

Are there different or better ways to meet the needs while designing out the substances of concern? Brainstorm to come up with new ideas using the Material Health Strategy Cards

Describe and illustrate 3 concepts for improvements by using the Material Health Strategy Cards. Try to be visual and give it a memorable name.

CONCEPT IDEA

CONCEPT 1: Biodegradable Resin PCB

A lighting system using biodegradable resin-based PCB substrates instead of conventional epoxy materials. Traditional epoxy resins make PCB separation and recycling difficult at end-of-life due to their permanent and non-degradable structure. By introducing biodegradable resin alternatives, the bulb becomes easier to disassemble and its electronic components can be separated more efficiently during recycling processes. This approach supports circular product design by reducing toxic material use and improving material recovery while maintaining the functionality of the electronic system.

CONCEPT 2: Glue-Free Service Bulb

A glue-free bulb designed with mechanical connections: clips instead of adhesive. This directly eliminates epoxy resin, a toxic and non-biodegradable substance that prevents PCB separation and recycling. The bulb can be fully disassembled, and components are collected, repaired or recycled through a take-back system. Removing epoxy resin also makes it possible to recover copper and aluminium from the PCB at end of life.

CONCEPT 3: Smart Diagnostic Light

A bulb with an integrated failure detection system that identifies whether the LED plate or the PCB driver needs replacing. Since the PCB contains ethylene glycol in its capacitors, a toxic electrolyte, the goal is to replace it only when strictly necessary. A simple current bridge triggers an indicator light when the PCB is faulty, avoiding unnecessary replacement and reducing toxic waste over time.

CONCEPT 4: Gold-Free PCB Surface Finish

A bulb PCB designed with OSP (Organic Solderability Preservative) surface finish instead of the conventional ENIG (Electroless Nickel Immersion Gold) plating. Gold, despite its negligible mass (0.000006 kg per bulb), carries an extremely high carbon footprint of 3767 kgCO₂e/kg, making it one of the most environmentally damaging materials per gram in the entire product. By switching to OSP, developed in-house by LumiNova, the process eliminates gold and nickel plating entirely. OSP is non-toxic, biodegradable and fully compatible with the bio-based PCB substrate. This change also simplifies end-of-life recovery, as the absence of noble metal plating makes copper extraction from recycled PCBs significantly cleaner.

REFLECT (10 MINS)

WHAT NEEDS TO BE INCLUDED IN THE DESIGN PROCESS?

What data would be important to have? What infrastructure is needed?
Who would you need to collaborate with?

Designing a Modular Lighting System

Designing a modular lighting system requires considering the full lifecycle of each component. In most LED bulbs, the driver PCB is typically the first component to fail because of heat and electrical stress, while the LED board can continue functioning for much longer. For this reason, the product is designed with replaceable modules, allowing users to change only the faulty driver instead of discarding the entire bulb. This approach extends product lifespan and significantly reduces electronic waste.

Material selection is also essential in creating a circular lighting product. The design replaces conventional epoxy-based PCB materials with biodegradable resin alternatives in order to facilitate disassembly at end-of-life. Using recyclable aluminium housings and glue-free mechanical connections further improves component separation and repairability. Replacing permanent adhesives with clips makes it easier to recover materials such as aluminium and electronic parts during recycling processes.

To support modularity at a larger scale, new infrastructure and services would be required. Repair and take-back systems could allow consumers to return defective modules for refurbishment or recycling. Manufacturers may also need adapted production lines capable of assembling interchangeable components instead of sealed single-use products. In addition, digital tracking systems such as QR codes could provide repair instructions, material information and maintenance history for each module.

Collaboration between designers, electronics manufacturers, recyclers and retailers is fundamental to making this system viable. Material experts can help identify safer and biodegradable material alternatives, while recycling companies ensure valuable materials re-enter production cycles. Through this collaborative and modular approach, the lighting industry can reduce waste generation, lower resource consumption and move towards a more circular product ecosystem.

Figure S2: Safe & Circular Product Redesign Workshop (continued)

Item No.	Level	Item title	Description	Material	Weight mass	Volume mass	CO2e/kgCO2e/pull-up kgCO2e	Manufacturing method (LCA style)	Supplier model	Shipping detail	Source		
1.0	0		Plastics LED A60 E27 bulb, SKU	Mixed	0.0000	0.0300	-	0	1.742064	Final assembly + electrical test + pack-o-Signify China final assembly/Consumer SKU assembled in China; downstream	EPD Signify Signify Catalogue		
1.1	1.0		Philips optical envelope assembly	Polymer	0.0000	0.0125	-	0	0.10695	Optical dome molding + bonding + interAPAC plastics supplier → Ch500 km road supplier → CN plant	PlasticsEurope ICE v3		
1.1.1	2.1		Outer diffuser dome	Polycarbonate / diffuser polymer	0.0085	0.0085	3.5	0.02975	0.08755	Injection molding + surface frosting	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope
1.1.1.1	3.1.1.1		Diffuser polymer feed	Polycarbonate family	0.0000	0.0085	-	0	0.0578	Polymer feed preparation	Integrated upstream	Included upstream	PlasticsEurope
1.1.1.1.1	4.1.1.1.1		Base polymer fraction	PC-like fossil polymer	0.0078	0.0078	6.8	0.05304	0.05304	Polymerization	Polymer supplier	Upstream integrated	ICE v3 PlasticsEurope
1.1.1.1.2	4.1.1.1.2		Diffusion / opacity additives	Mineral filler / additive package	0.0007	0.0007	6.8	0.00476	0.00476	Compounding	Additive supplier	Upstream integrated	ICE v3
1.1.2	2.1		Inner optical shell / mixing layer	Diffuser polymer	0.0015	0.0015	6.8	0.0102	0.0102	Thin-wall injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope
1.1.3	2.1		Neck bonding / seal	Epoxy adhesive	0.0005	0.0005	6	0.003	0.003	Adhesive dispensing + curing	APAC chemical supplier	500 km road to CN assembly	ICE v3
1.1.4	2.1		LED board plastic support ring	Plastic "4" observed	0.0012	0.0012	3.1	0.00372	0.00372	Injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope
1.1.5	2.1		Driver PCB plastic holder / inner	Plastic "4" observed	0.0008	0.0008	3.1	0.00248	0.00248	Injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope
1.2	1.0		Thermal / structural module	Metal + polymer	0.0000	0.0047	-	0	0.09507	Internal heat spreading + structural support	APAC supplier → China plant	500 km road supplier → CN plant	EPD Signify Hischer 2015
1.2.1	2.1.2		Internal heat spreader / metal	Aluminium alloy	0.0035	0.0035	8.5	0.02975	0.09135	Stamping / deep drawing / forming	APAC metal supplier	500 km road to CN assembly	ICE v3 EU Aluminium 2022
1.2.1.1	3.1.2.1		Aluminium stock	Aluminium sheet / cup stock	0.0000	0.0035	-	0	0.0616	Sheet preparation	Integrated upstream	Included upstream	ICE v3
1.2.1.1.1	4.1.2.1.1		Primary aluminium share	Aluminium	0.0033	0.0033	17.6	0.05808	0.05808	Smelting + rolling	Aluminium supplier	Upstream integrated	ICE v3 EU Aluminium 2022
1.2.1.1.2	4.1.2.1.2		Alloying fraction	Si / Mg alloying	0.0002	0.0002	17.6	0.00352	0.00352	Alloying	Alloy supplier	Upstream integrated	ICE v3 Ecoinvent 3
1.2.2	2.1.2		Secondary internal support	Plastic "4" observed	0.0012	0.0012	3.1	0.00372	0.00372	Injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope
1.3	1.0		LED light engine assembly	Electronics + polymer	0.0000	0.0043	-	0	0.525036	LED board + optical interface + locator	APAC LED supplier → China	800 km road supplier → CN plant	EPD Signify Hischer 2015
1.3.1	2.1.3		LED board SHPC-Z JXL 4421	MCPBC + LEDs	0.0022	0.0022	15.1	0.033198	0.496601	MCPBC fabrication + SMT LED mounting	APAC LED board supplier	800 km road to CN assembly	Ecoinvent 3 Hischer 2015
1.3.1.1	3.1.3.1		MCPBC substrate share	Aluminium + dielectric + copper	0.0012	0.0012	10.7	0.012804	0.360632	Aluminium-core PCB manufacture	Integrated upstream	Included upstream	ICE v3 Hischer 2015
1.3.1.1.1	4.1.3.1.1		Aluminium core	Aluminium	0.00078	0.00078	17.6	0.013728	0.013728	Sheet rolling / punching	Metal supplier	Upstream integrated	ICE v3 EU Aluminium 2022
1.3.1.1.2	4.1.3.1.2		Dielectric layer	Thermally conductive dielectric	0.00010	0.00010	6	0.0006	0.0006	Coating / lamination	PCB supplier	Upstream integrated	Ecoinvent 3
1.3.1.1.3	4.1.3.1.3		Copper circuit layer	Copper foil	0.00029	0.00029	32	0.00928	0.00928	Copper lamination + etching	PCB supplier	Upstream integrated	Ecoinvent 3 Lenz 2021
1.3.1.1.4	4.1.3.1.4		Nickel finish trace	Nickel	0.00002	0.00002	11	0.00022	0.00022	Surface finishing / plating	PCB supplier	Upstream integrated	ICE v3 ICE v3 Ecoinvent 3
1.3.1.1.5	4.1.3.1.5		Gold finish trace	Gold	0.00001	0.00010	32400	0.324	0.324	Surface finishing	PCB supplier	Upstream integrated	Ecoinvent 3 Ecoinvent 3 Lenz 2021
1.3.1.2	3.1.3.2		LED packages share	Semiconductor + package meta	0.00070	0.00070	25.9	0.018102	0.096572	LED package assembly	Integrated upstream	Included upstream	Ecoinvent 3 Hischer 2015
1.3.1.2.1	4.1.3.1.2		Semiconductor die share	Gallium high grade / semiconductor	0.00020	0.00020	200	0.04	0.04	Epitaxy + dicing	Semiconductor supplier	Upstream integrated	USGS Gallium 2025 Hischer 2015 Nishijima et al. gallium nitride LED car
1.3.1.2.2	4.1.3.1.2		Leadframe / package metals	Copper	0.00018	0.00018	32	0.00576	0.00576	Stamping + plating	Package supplier	Upstream integrated	Ecoinvent 3 ICE v3
1.3.1.2.3	4.1.3.1.2		Package resin / lens	Silicone / resin	0.00028	0.00028	6.5	0.00182	0.00182	Dispensing + curing	Package supplier	Upstream integrated	Ecoinvent 3 Hischer 2015
1.3.1.2.4	4.1.3.1.2		Silver interconnect trace	Silver	0.00003	0.00003	763	0.02289	0.02289	Wire bonding / metallization	Package supplier	Upstream integrated	Ecoinvent 3 ICE v3
1.3.1.2.5	4.1.3.1.2		Palladium trace	Palladium	0.00001	0.00001	800	0.008	0.008	Metallization / plating	Package supplier	Upstream integrated	Ecoinvent 3 Ecoinvent 3 USGS PGMs 2025
1.3.1.3	3.1.3.1		Solder / interface share	Sn-based solder + minor silver	0.00030	0.00030	7.3	0.002199	0.006199	Reflow soldering	Integrated upstream	Included upstream	ICE v3 Ecoinvent 3
1.3.1.3.1	4.1.3.1.3		Tin solder fraction	Tin	0.00028	0.00028	5	0.0014	0.0014	Solder alloy preparation	Integrated upstream	Included upstream	ICE v3 USGS Tin 2025
1.3.1.3.2	4.1.3.1.3		Silver solder fraction	Silver	0.00002	0.00002	130	0.0026	0.0026	Solder alloy preparation	Integrated upstream	Included upstream	Ecoinvent 3 ICE v3
1.3.2	2.1.3		Optical silicone / phosphor interconnect	Silicone + yttrium oxide	0.0010	0.0010	7.5	0.0075	0.025025	Coating / dispensing + cure	APAC LED supplier	800 km road to CN assembly	Ecoinvent 3 ICE v3
1.3.2.1	3.1.3.2		Optical silicone fraction	Silicone	0.00085	0.00085	6.5	0.005525	0.005525	Dispensing + cure	APAC LED supplier	Included upstream	Ecoinvent 3 ICE v3
1.3.2.2	3.1.3.2		Phosphor fraction	Yttrium oxide	0.00015	0.00015	80	0.012	0.012	Phosphor mixing / coating	Chemical supplier	Included upstream	USGS REE 2025 Sreeracher 2014 Yttrium oxide LCA 10
1.3.3	2.1.3		LED board plastic holder	Plastic "4" observed	0.0011	0.0011	3.1	0.00341	0.00341	Injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope
1.4	1.0		Driver electronics assembly	Electronics	0.0000	0.0048	-	0	0.955708	Driver PCB + capacitor + plastic guide	APAC electronics supplier →	800 km road supplier → CN plant	EPD Signify Hischer 2015
1.4.1	2.1.4		Driver PCB SHPC-Z JXL 3086	FR4 PCB + SMT parts	0.0024	0.0024	13.5	0.0324	0.884568	FR4 fabrication + SMT assembly	APAC PCB supplier	800 km road to CN assembly	Ecoinvent 3 Hischer 2015
1.4.1.1	3.1.4.1		FR4 laminate share	Glass fiber + epoxy + copper	0.00110	0.00110	13.2	0.014498	0.676628	Lamination + copper cladding + etching	Integrated upstream	Included upstream	ICE v3 Ecoinvent 3
1.4.1.1.1	4.1.4.1.1		Glass fiber fraction	Glass fiber	0.00050	0.00050	8	0.004	0.004	Fiber production	Materials supplier	Upstream integrated	ICE v3 ICE v3 Ecoinvent 3
1.4.1.1.2	4.1.4.1.1		Epoxy fraction	Epoxy resin	0.00030	0.00030	6	0.0018	0.0018	Resin synthesis	Chemical supplier	Upstream integrated	ICE v3 Hammond & Jones ICE v3.0 table 5.1
1.4.1.1.3	4.1.4.1.1		Copper layer	Copper foil	0.00025	0.00025	32	0.008	0.008	Copper cladding	Metal supplier	Upstream integrated	Ecoinvent 3 ICE v3
1.4.1.1.4	4.1.4.1.1		Nickel finish trace	Nickel	0.00003	0.00003	11	0.00033	0.00033	Surface finishing / plating	PCB supplier	Upstream integrated	ICE v3 ICE v3 Ecoinvent 3
1.4.1.1.5	4.1.4.1.1		Gold finish trace	Gold	0.00002	0.00002	32400	0.648	0.648	Surface finishing	PCB supplier	Upstream integrated	Ecoinvent 3 Ecoinvent 3 Lenz 2021
1.4.1.2	3.1.4.1.2		Semiconductor / rectifier / contr	Silicon + mold compound + tra	0.00042	0.00042	26.19	0.011	0.11789	Semiconductor packaging	Integrated upstream	Included upstream	Ecoinvent 3 Williams 2004
1.4.1.2.1	4.1.4.1.2		Silicon / mold compound fraction	Silicon + mold compound	0.00038	0.00038	200	0.076	0.076	Semiconductor packaging	Integrated upstream	Included upstream	Ecoinvent 3 Williams 2004
1.4.1.2.2	4.1.4.1.2		Silver contact trace	Silver	0.00003	0.00003	763	0.02289	0.02289	Metallization	Integrated upstream	Included upstream	Ecoinvent 3 ICE v3
1.4.1.2.3	4.1.4.1.2		Palladium trace	Palladium	0.00001	0.00001	800	0.008	0.008	Metallization	Integrated upstream	Included upstream	Ecoinvent 3 Ecoinvent 3 USGS PGMs 2025
1.4.1.3	3.1.4.1.3		Passive component share	Ceramic + metals	0.00050	0.00050	11.4	0.0057	0.04955	Passive component manufacture	Integrated upstream	Included upstream	Ecoinvent 3 Hischer 2015
1.4.1.3.1	4.1.4.1.3		Ceramic body share	Ceramic	0.00030	0.00030	3	0.0009	0.0009	Passive component manufacture	Integrated upstream	Included upstream	ICE v3 Ecoinvent 3
1.4.1.3.2	4.1.4.1.3		Base metal share	Nickel / copper	0.00015	0.00015	32	0.0048	0.0048	Metallization / forming	Integrated upstream	Included upstream	ICE v3 Ecoinvent 3
1.4.1.3.3	4.1.4.1.3		Silver termination trace	Silver	0.00005	0.00005	763	0.03815	0.03815	Termination metallization	Integrated upstream	Included upstream	Ecoinvent 3 ICE v3
1.4.1.4	3.1.4.1.4		Solder / flux share	Solder + flux + minor silver	0.00038	0.00038	6.58	0.0025	0.0081	Reflow soldering	Integrated upstream	Included upstream	ICE v3 Ecoinvent 3
1.4.1.4.1	4.1.4.1.4		Tin solder fraction	Tin	0.00033	0.00033	5	0.00165	0.00165	Solder alloy preparation	Integrated upstream	Included upstream	ICE v3 USGS Tin 2025
1.4.1.4.2	4.1.4.1.4		Silver solder fraction	Silver	0.00003	0.00003	130	0.0039	0.0039	Solder alloy preparation	Integrated upstream	Included upstream	Ecoinvent 3 ICE v3
1.4.1.4.3	4.1.4.1.4		Flux / organics fraction	Flux / organics	0.00002	0.00002	2.5	0.00005	0.00005	Flux formulation	Integrated upstream	Included upstream	Ecoinvent 3
1.4.2	2.1.4		Electrolytic capacitor	Aluminium + electrolyte + pap	0.0010	0.0010	18	0.018	0.0379	Capacitor winding + filling + sealing	APAC capacitor supplier	800 km road to CN assembly	Ecoinvent 3 Hischer 2015
1.4.2.1	3.1.4.2		Aluminium can / foil share	Aluminium	0.0005	0.0005	17.6	0.0088	0.0088	Foil rolling + can forming	Capacitor supplier	Upstream integrated	Ecoinvent 3
1.4.2.2	3.1.4.2		Electrolyte share	Chemical electrolyte	0.0002	0.0002	50	0.01	0.01	Chemical formulation	Chemical supplier	Upstream integrated	Ecoinvent 3
1.4.2.3	3.1.4.2		Paper separator share	Cellulosic paper	0.0001	0.0001	1	0.0001	0.0001	Paper conversion	Paper supplier	Upstream integrated	Ecoinvent 3
1.4.2.4	3.1.4.2		Rubber / seal share	Elastomer	0.0002	0.0002	5	0.001	0.001	Elastomer molding	Polymer supplier	Upstream integrated	Ecoinvent 3 ICE v3
1.4.3	2.1.4		Driver housing / plastic guide	Plastic "4" observed	0.0014	0.0014	3.1	0.00434	0.00434	Injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope
1.4.4	2.1.4		Additional driver overheat / as	Electricity, auxiliaries, scrap.	0.0000	0.0000	-	0.0289	0.0289	SMT engine + test + process losses	Signify / supply-chain overh	Modeled as plant / process burden	ICE 60238 standard E27 cap architecture ICE v3
1.5	1.0		E27 cap assembly	Metal + ceramic	0.0000	0.0022	-	0	0.03574	Cap forming + contact assembly	Standard E27 supplier → Ch500 km road supplier → CN plant	ICE 60238 standard E27 cap architecture ICE v3	
1.5.1	2.1.5		Screen shell E27	Aluminium / thin metal shell	0.0014	0.0014	17.6	0.02464	0.02464	Stamping + forming	APAC cap supplier	500 km road to CN assembly	ICE v3 EU Aluminium 2022
1.5.2	2.1.5		Bottom contact	Copper alloy	0.0003	0.0003	32	0.0096	0.0096	Stamping	APAC cap supplier	500 km road to CN assembly	ICE v3 Ecoinvent 3
1.5.3	2.1.5		Electrical insulator	Ceramic / thermoset	0.0005	0.0005	3	0.0015	0.0015	Pressing + sintering / molding	APAC ceramic supplier	500 km road to CN assembly	ICE v3 Ecoinvent 3
1.6	1.0		Internal wiring	Metal	0.0000	0.0007	-	0	0.0224	Wire cutting + soldering	Standard bulb architecture	300 km road supplier → CN plant	ICE v3
1.6.1	2.1.6		Internal wires	Copper / tinned copper	0.0007	0.0007	32	0.0224	0.0224	Wire drawing + cutting	APAC wire supplier	300 km road to CN assembly	ICE v3 Ecoinvent 3
1.7	1.0		Packaging allocation per bulb	Paper + PE	0.0000	0.0008	-	0	0.00116	Carton conversion + film extrusion	Consumer pack allocation	p100 km road supplier → CN plant	EPD Signify ICE v3
1.7.1	2.1.7		Cardboard share per bulb	Paperboard	0.0006	0.0006	0.9	0.00054	0.00054	Die cutting + printing	Packaging supplier	100 km road to CN assembly	ICE v3 Ecoinvent 3
1.7.2	2.1.7		Plastic film share per bulb	PE	0.0002	0.0002	3.1	0.00062	0.00062	Film extrusion	Packaging supplier	100 km road to CN assembly	ICE v3 PlasticsEurope

Figure S3: Bill of material of the incumbent Philips Light Bulb

Item No.	Level	Parent Item	Description	Material	Important				Important				Manufacturing method (LCA style)	Supplier model	Shipping detail		Source
					Direct mass (kg)	Roll-up mass (kg)	kgCO2e/kg	kgCO2e/part	Roll-up kgCO2e	Direct mass (kg)	Roll-up mass (kg)	kgCO2e/kg			Consumer SKU assembled in	China final assembly	
1.0	0		Philips LED A60 E27 bulb, SKU 87201	Mixed	0.0000	0.0025	-	0	0.0000	0.0000	0.0000	-	Final assembly + electrical test + pack-out	Signify	China final assembly	China final assembly	ChrEPD Signify Signify catalogue
1.1	11.0		Plastic optical envelope assembly	Polymer	0.0000	0.0120	-	0	0.0500	0.0000	0.0120	-	Optical dome molding + bonding + internal	APAC plastics supplier	500 km road to CN assembly	PlasticsEurope ICE v3	
1.1.1	21.1		Outer diffuser dome	Polycarbonate / diffuser poly	0.0005	0.0100	3.5	0.02975	0.0400	0.0000	0.0100	3.5	Injection molding + surface frosting	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope	
1.1.1.1	31.1.1		Diffuser polymer feed	Polycarbonate family	0.0000	0.0085	-	0	0.0102	0.0000	0.0085	-	Polymer feed preparation	Integrated upstream	Included upstream	PlasticsEurope	
1.1.1.1.1	41.1.1.1		Base polymer fraction	PC-like silsil polymer	0.0078	0.0078	1.2	0.00936	0.00936	0.0000	0.0078	1.2	Polymerization	Polymer supplier	Upstream integrated	ICE v3 PlasticsEurope	
1.1.1.1.2	41.1.1.1.2		Diffusion / opacity additives	Mineral filler / additive pack	0.0007	0.0007	1.2	0.00084	0.00084	0.0000	0.0007	1.2	Additive compounding	Compounding	Upstream integrated	ICE v3	
1.1.2	21.1		Inner optical shell / mixing layer	Diffuser polymer	0.0015	0.0015	1.2	0.0018	0.0018	0.0000	0.0015	1.2	Thin-wall injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope	
1.1.3	21.1		Neck bonding / seal	Epoxy adhesive	0	0	4	0.002	0.002	0.0000	0	4	Adhesive dispensing + curing	APAC chemical supplier	500 km road to CN assembly	ICE v3	
1.1.4	21.1		LED board plastic support ring	Plastic "4" observed	0.0012	0.0012	3.1	0.00372	0.00372	0.0000	0.0012	3.1	Injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope	
1.1.5	21.1		Driver PCB plastic holder / inner chassis	Plastic "4" observed	0.0008	0.0008	3.1	0.00248	0.00248	0.0000	0.0008	3.1	Injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope	
1.2	11.0		Thermal / structural module	Metal + polymer	0.0000	0.0047	0	0.04152	0	0.0000	0.0047	0	Internal heat spreading + structural support	APAC supplier → China plant	500 km road supplier → CN plant	EPD Signify Hirschler 2015	
1.2.1	21.2		Internal heat spreader / metal cup	Aluminium alloy	0.0035	0.0035	8.5	0.02975	0.0378	0.0000	0.0035	8.5	Stamping / deep drawing / forming	APAC metal supplier	500 km road to CN assembly	ICE v3 EU Aluminium	
1.2.1.1	31.2.1		Aluminium stock	Aluminium sheet / cup stock	0.0000	0.0035	-	0	0.00805	0.0000	0.0035	-	Sheet preparation	Integrated upstream	Included upstream	ICE v3	
1.2.1.1.1	41.2.1.1		Recycled aluminium share	Recycled aluminium	0.0033	0.0033	2.3	0.00759	0.00759	0.0000	0.0033	2.3	Secondary aluminium smelting + rolling	Aluminium recycler	Upstream integrated	ICE v3 EU Aluminium	
1.2.1.1.2	41.2.1.1.2		Alloying fraction	Si / Mg alloying	0.0002	0.0002	2.3	0.00046	0.00046	0.0000	0.0002	2.3	Alloying	Alloy supplier	Upstream integrated	ICE v3 Ecoinvent 3	
1.2.2	21.2		Secondary internal support	Plastic "4" observed	0.0012	0.0012	3.1	0.00372	0.00372	0.0000	0.0012	3.1	Injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope	
1.3	11.0		LED light engine assembly	Electronics + polymer	0.0000	0.00429	-	0	0.206962	0.0000	0.00429	-	LED board + optical interface + locator	APAC LED supplier → China	pi800 km road supplier → CN plant	EPD Signify Hirschler 2015	
1.3.1	21.3		LED board SHPC-Z JXL 4421	MCPCB + LEDs	0.0022	0.00219	15.09	0.033198	0.184527	0.0000	0.0022	15.09	MCPCB fabrication + SMT LED mounting	APAC LED board supplier	800 km road to CN assembly	Ecoinvent 3 Hirschler 2015	
1.3.1.1	31.3.1		MCPCB substrate share	Aluminium + dielectric + copo	0.0012	0.0012	10.67	0.012804	0.062068	0.0000	0.0012	10.67	Aluminium-core PCB manufacture	Integrated upstream	Included upstream	Ecoinvent 3 Hirschler 2015	
1.3.1.1.1	41.3.1.1.1		Aluminium core (recycled)	Recycled aluminium	0.00078	0.00078	2.3	0.001794	0.001794	0.0000	0.00078	2.3	Sheet rolling / punching	Metal supplier	Upstream integrated	ICE v3 EU Aluminium	
1.3.1.1.2	41.3.1.1.2		Bio-based dielectric layer	Thermally conductive bio-resin	0.00010	0.0001	4	0.0004	0.0004	0.0000	0.00010	4	Coating / lamination	PCB supplier	Upstream integrated	Ecoinvent 3	
1.3.1.1.3	41.3.1.1.3		Copper circuit layer	Copper foil	0.00029	0.00029	32	0.00028	0.00028	0.0000	0.00029	32	Copper lamination + etching	PCB supplier	Upstream integrated	Ecoinvent 3 Lenz 2021	
1.3.1.1.4	41.3.1.1.4		Nickel finish trace	Nickel	0.00002	0.00002	6	0.00012	0.00012	0.0000	0.00002	6	Surface finishing / plating	PCB supplier	Upstream integrated	refined electroplati Ecoinvent 3	
1.3.1.1.5	41.3.1.1.5		Reduced gold finish trace	Gold	0.000002	0.000002	3767	0.03767	0.03767	0.0000	0.000002	3767	Surface finishing	PCB supplier	Upstream integrated	ENIG trace → 40 kgCO2 LenZ 2021	
1.3.1.1.6	41.3.1.1.6		Copper finish share	Copper	0.000004	0.000004	25.86	0.018102	0.084062	0.0000	0.000004	25.86	Surface finishing	PCB supplier	Upstream integrated	Ecoinvent 3	
1.3.1.1.7	41.3.1.1.7		Silver finish share	Silver	0.000004	0.000004	200	0.04	0.04	0.0000	0.000004	200	Surface finishing	PCB supplier	Upstream integrated	Ecoinvent 3 ICE v3	
1.3.1.2	31.3.1		LED packages share	Semiconductor + package mo	0.00070	0.00069	32	0.00576	0.00576	0.0000	0.00070	32	LED package assembly	Integrated upstream	Included upstream	Ecoinvent 3 Hirschler 2015	
1.3.1.2.1	41.3.1.2.1		Semiconductor die share	Gallium high grade / semicon	0.00020	0.0002	6.5	0.00182	0.00182	0.0000	0.00020	6.5	Epitaxy + dicing	Semiconductor supplier	Upstream integrated	USGS Gallium Hirschler 2015	
1.3.1.2.2	41.3.1.2.2		Leadframe / package metals	Copper	0.00018	0.00018	446	0.00338	0.01338	0.0000	0.00018	446	Stamping + plating	Package supplier	Upstream integrated	Ecoinvent 3 ICE v3	
1.3.1.2.3	41.3.1.2.3		Package resin / lens	Silicone / resin	0.00028	0.00028	500	0.005	0.005	0.0000	0.00028	500	Dispensing + curing	Package supplier	Upstream integrated	Ecoinvent 3 Hirschler 2015	
1.3.1.2.4	41.3.1.2.4		Silver interconnect trace	Silver	0.00003	0.00003	7.33	0.002199	0.005199	0.0000	0.00003	7.33	Wire bonding / metallization	Package supplier	Upstream integrated	Ecoinvent 3 ICE v3	
1.3.1.3	31.3.1		Solder / interface share	Sn-based solder + minor silv	0.00030	0.0003	5	0.0014	0.0014	0.0000	0.00030	5	Reflow soldering	Integrated upstream	Included upstream	ICE v3 Ecoinvent 3	
1.3.1.3.1	41.3.1.3.1		Tin solder fraction	Tin	0.00028	0.00028	80	0.0016	0.0016	0.0000	0.00028	80	Solder alloy preparation	Integrated upstream	Included upstream	ICE v3 USGS Tin	
1.3.1.3.2	41.3.1.3.2		Optical silicone / phosphor interface	Silicone + yttrium oxide	0.0010	0.001	6.5	0.005525	0.005525	0.0000	0.0010	6.5	Coating / dispensing + cure	APAC LED supplier	800 km road to CN assembly	Ecoinvent 3 ICE v3	
1.3.2	21.3		Optical silicone fraction	Silicone	0.00085	0.00085	40	0.006	0.006	0.0000	0.00085	40	Dispensing + cure	APAC LED supplier	Included upstream	Ecoinvent 3 ICE v3	
1.3.2.1	31.3.2		Phosphor fraction	Yttrium oxide	0.00015	0.00015	3.1	0.00341	0.00341	0.0000	0.00015	3.1	Phosphor mixing / coating	Chemical supplier	Included upstream	USGS REF Sprecher 2014	
1.3.2.2	31.3.2		LED board plastic locator	Plastic "4" observed	0.0011	0.0011	-	0	0.272688	0.0000	0.0011	-	Injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope	
1.4	11.0		Driver electronics assembly	Electronics	0.0000	0.00479	13.5	0.0324	0.281298	0.0000	0.00479	13.5	Driver PCB + capacitor + plastic guide +	APAC electronics supplier →	800 km road supplier → CN plant	EPD Signify Hirschler 2015	
1.4.1	21.4		Driver PCB SHPC-Z JXL 3086	FR4 PCB + SMT parts	0.0024	0.00239	13.18	0.014498	0.103218	0.0000	0.0024	13.18	FR4 fabrication + SMT assembly	APAC PCB supplier	800 km road to CN assembly	Ecoinvent 3 Hirschler 2015	
1.4.1.1	31.4.1		FR4 laminate share	Glass fiber + epoxy + copper	0.00110	0.0011	8	0.004	0.004	0.0000	0.00110	8	Lamination + copper cladding + etching	Integrated upstream	Included upstream	ICE v3 Ecoinvent 3	
1.4.1.1.1	41.4.1.1		Glass fiber fraction	Glass fiber	0.00050	0.0005	4	0.0012	0.0012	0.0000	0.00050	4	Fiber production	Materials supplier	Upstream integrated	PCB-grade woven → 10 Ecoinvent 3	
1.4.1.1.2	41.4.1.1.2		Biodegradable resin fraction	Bio-based epoxy / biodegrad	0.00030	0.0003	32	0.008	0.008	0.0000	0.00030	32	Bio-resin synthesis	Chemical supplier	Upstream integrated	Ecoinvent 3 PlasticsEurope	
1.4.1.1.3	41.4.1.1.3		Copper layer	Copper foil	0.00025	0.00025	6	0.00018	0.00018	0.0000	0.00025	6	Copper cladding	Metal supplier	Upstream integrated	Ecoinvent 3 ICE v3	
1.4.1.1.4	41.4.1.1.4		Nickel finish trace	Nickel	0.00003	0.00003	3767	0.07534	0.07534	0.0000	0.00003	3767	Surface finishing / plating	PCB supplier	Upstream integrated	refined electroplati Ecoinvent 3	
1.4.1.1.5	41.4.1.1.5		Reduced gold finish trace	Gold	0.000004	0.000004	26.19	0.011	0.10538	0.0000	0.000004	26.19	Surface finishing	PCB supplier	Upstream integrated	ENIG trace → 40 kgCO2 LenZ 2021	
1.4.1.1.6	41.4.1.1.6		Copper finish share	Copper	0.000008	0.000008	200	0.076	0.076	0.0000	0.000008	200	Surface finishing	PCB supplier	Upstream integrated	Ecoinvent 3	
1.4.1.1.7	41.4.1.1.7		Silver finish share	Silver	0.000008	0.000008	446	0.01338	0.01338	0.0000	0.000008	446	Surface finishing	PCB supplier	Upstream integrated	Ecoinvent 3 ICE v3	
1.4.1.2	31.4.1		Semiconductor / rectifier / control IC s	Silicon + mold compound +	0.00042	0.00041	500	0.005	0.005	0.0000	0.00042	500	Semiconductor packaging	Integrated upstream	Included upstream	Ecoinvent 3 Williams 2004	
1.4.1.2.1	41.4.1.2		Silicon / mold compound fraction	Silicon + mold compound	0.00038	0.00038	11.4	0.0057	0.0337	0.0000	0.00038	11.4	Semiconductor packaging	Integrated upstream	Included upstream	Ecoinvent 3 Williams 2004	
1.4.1.2.2	41.4.1.2.2		Silver contact trace	Silver	0.00003	0.00003	3	0.0009	0.0009	0.0000	0.00003	3	Metallization	Integrated upstream	Included upstream	ICE v3 Ecoinvent 3	
1.4.1.3	31.4.1		Passive component share	Ceramic + metals	0.00050	0.0005	32	0.0048	0.0048	0.0000	0.00050	32	Passive component manufacture	Integrated upstream	Included upstream	Ecoinvent 3 Hirschler 2015	
1.4.1.3.1	41.4.1.3.1		Ceramic body share	Ceramic	0.00030	0.0003	446	0.0223	0.0223	0.0000	0.00030	446	Passive component manufacture	Integrated upstream	Included upstream	ICE v3 Ecoinvent 3	
1.4.1.3.2	41.4.1.3.2		Base metal share	Nickel / copper	0.00015	0.00015	6.58	0.0025	0.0066	0.0000	0.00015	6.58	Metallization / forming	Integrated upstream	Included upstream	ICE v3 Ecoinvent 3	
1.4.1.3.3	41.4.1.3.3		Silver termination trace	Silver	0.00005	0.00005	5	0.00165	0.00165	0.0000	0.00005	5	Termination metallization	Integrated upstream	Included upstream	Ecoinvent 3 ICE v3	
1.4.1.4	31.4.1		Solder / flux share	Solder + flux + minor silver	0.00038	0.00038	80	0.0024	0.0024	0.0000	0.00038	80	Reflow soldering	Integrated upstream	Included upstream	ICE v3 Ecoinvent 3	
1.4.1.4.1	41.4.1.4.1		Tin solder fraction	Tin	0.00033	0.00033	2.5	0.00005	0.00005	0.0000	0.00033	2.5	Solder alloy preparation	Integrated upstream	Included upstream	ICE v3 USGS Tin	
1.4.1.4.2	41.4.1.4.2		Silver solder fraction	Silver	0.00003	0.00003	18	0.018	0.0225	0.0000	0.00003	18	Solder alloy preparation	Integrated upstream	Included upstream	ICE v3 ICE v3	
1.4.1.4.3	41.4.1.4.3		Flux / organics fraction	Flux / organics	0.00002	0.00002	2.3	0.00115	0.00115	0.0000	0.00002	2.3	Flux formulation	Integrated upstream	Included upstream	Ecoinvent 3	
1.4.2	21.4		Electrolytic capacitor	Aluminium + electrolyte + pa	0.0010	0.001	10	0.002	0.002	0.0000	0.0010	10	Capacitor winding + filling + sealing	APAC capacitor supplier	800 km road to CN assembly	Ecoinvent 3 Hirschler 2015	
1.4.2.1	31.4.2		Aluminium can / foil share (recycled)	Recycled aluminium	0.0005	0.0005	1	0.0001	0.0001	0.0000	0.0005	1	Foil rolling + can forming	Capacitor supplier	Upstream integrated	ICE v3 Ecoinvent 3	
1.4.2.2	31.4.2		Electrolyte share	Chemical electrolyte	0.0002	0.0002	5	0.001	0.001	0.0000	0.0002	5	Chemical formulation	Chemical supplier	Upstream integrated	Ecoinvent 3	
1.4.2.3	31.4.2		Paper separator share	Cellulosic paper	0.0001	0.0001	3.1	0.00434	0.00434	0.0000	0.0001	3.1	Paper converting	Paper supplier	Upstream integrated	Ecoinvent 3	
1.4.2.4	31.4.2		Rubber / seal share	Elastomer	0.0002	0.0002	-	-0.0352	-0.0352	0.0000	0.0002	-	Elastomer molding	Polymer supplier	Upstream integrated	Ecoinvent 3 ICE v3	
1.4.3	21.4		Driver housing / plastic guide	Plastic "4" observed	0.0014	0.0014	-	0	0.01432	0.0000	0.0014	-	Injection molding	APAC plastics supplier	500 km road to CN assembly	ICE v3 PlasticsEurope	
1.4.4	21.4		Additional driver overhead / assembly	Electricity, auxiliaries, scrap	0.0000	0	2.3	0.00322	0.00322	0.0000	0	2.3	SMT energy + test + process losses	Signify / supply-chain over the	Modeled as plant / process burden	ICE v3	
1.5	11.0		E27 cap assembly	Metal + ceramic	0.0000	0.0022	32	0.0096	0.0096	0.0000	0.0022	32	Cap forming + contact assembly	Standard E27 supplier →	China500 km road supplier → CN plant	ICE v3	
1.5.1	21.5		Screw shell E27 (recycled aluminium)	Recycled aluminium	0.0014	0.0014	3	0.0015	0.0015	0.0000	0.0014	3	Stamping + forming	APAC cap supplier	500 km road to CN assembly	ICE v3 EU Aluminium	
1.5.2	21.5		Bottom contact	Copper alloy	0.0003	0.0003	-	0	0.0224	0.0000	0.0003	-	Stamping	APAC cap supplier	500 km road to CN assembly	ICE v3 Ecoinvent 3	
1.5.3																	

(1) scope 1&2, 3 upstream & downstream (EL)
(2) use/consumption

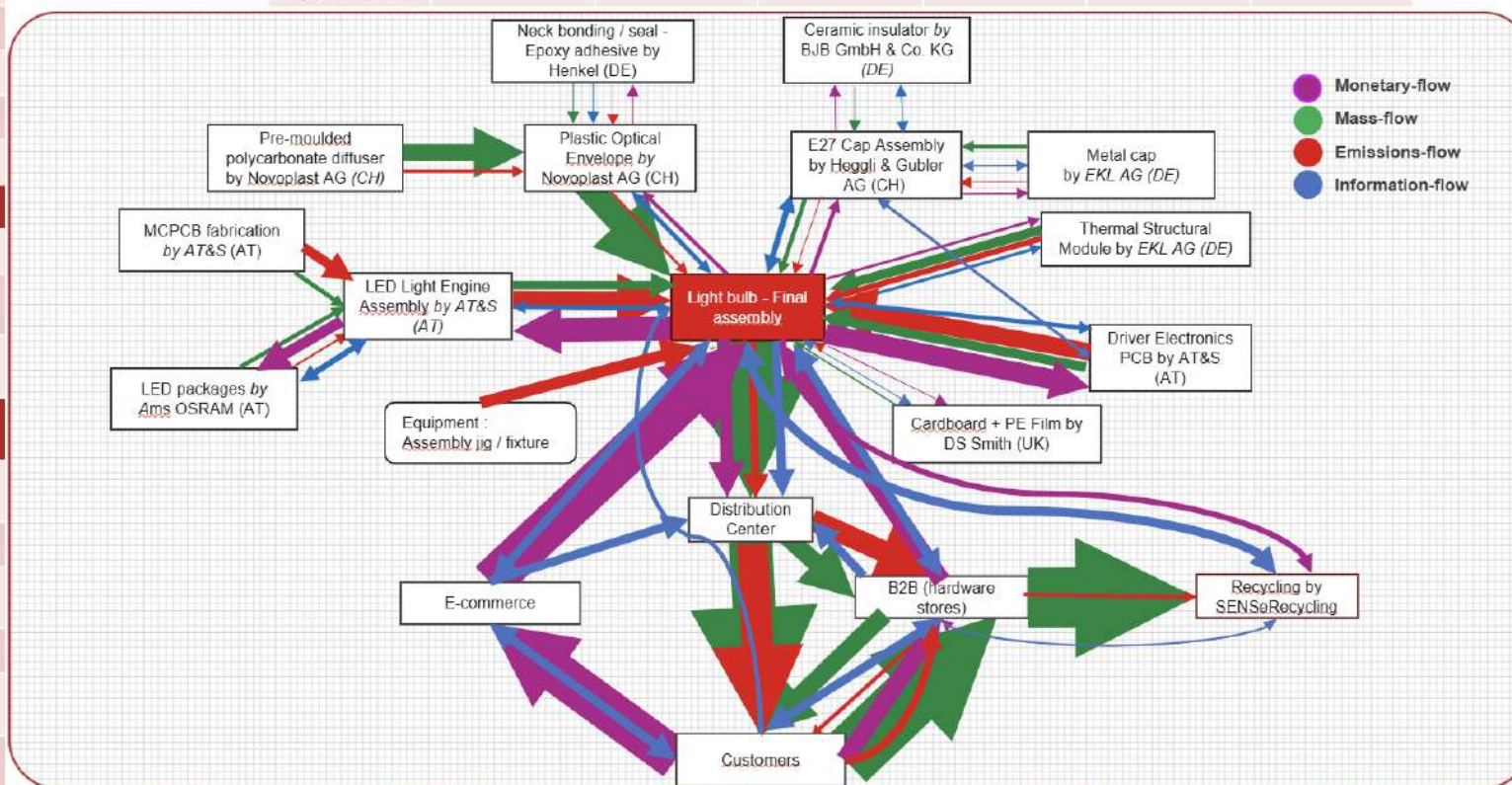
SDG	(1)	(2)
1 No Poverty		
2 Zero Hunger		
3 Good health		
4 Education		
5 Gender		
6 Clean water		
7 Clean energy		
8 Decent work		
9 Innovation / industry		
10 Reduced inequalities		
11 Cities and communities		
12 Consumption / production		
13 Climate		
14 Life below water		
15 Life on Land		
16 Peace & justice		
17 Partnerships for goals		

Which impacts do you reduce? (quantify)

	Climate change	Freshwater loss	Chemical & plastic pollution	Forest & seabed loss	Biodiversity loss	Soil nutrient pollution
Scope 1 direct						
Scope 2 indirect						
Scope 3 upstream						
Scope 3 downstream						

Where are you circular?

7 principles	
1 Longevity	
2 Reuse (refurbish / repair/ remanufacture)	
3 Resource-efficient manufacturing	
4 Recycling / design for disassembly	
5 Collaboration in value chain	
6 Local sourcing & production	
7 Business model innovation	



Specific	Measurable	Attainable	Relevant	Time-oriented	Slow	Close	Narrow	Regenerate

Figure S5: LumiNova's baseline linear supply chain scenario

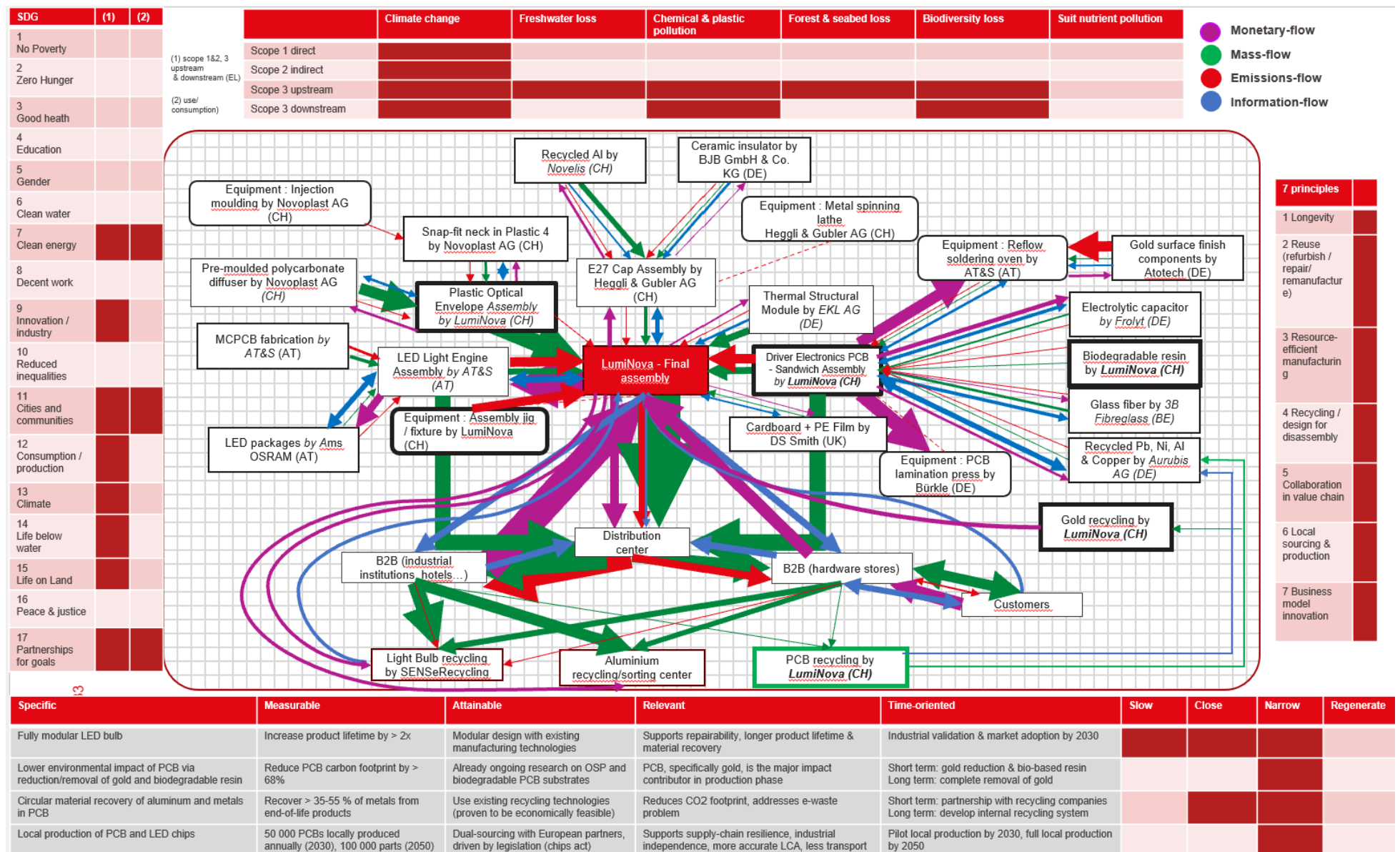


Figure S6: Circular Supply Chain - 2030

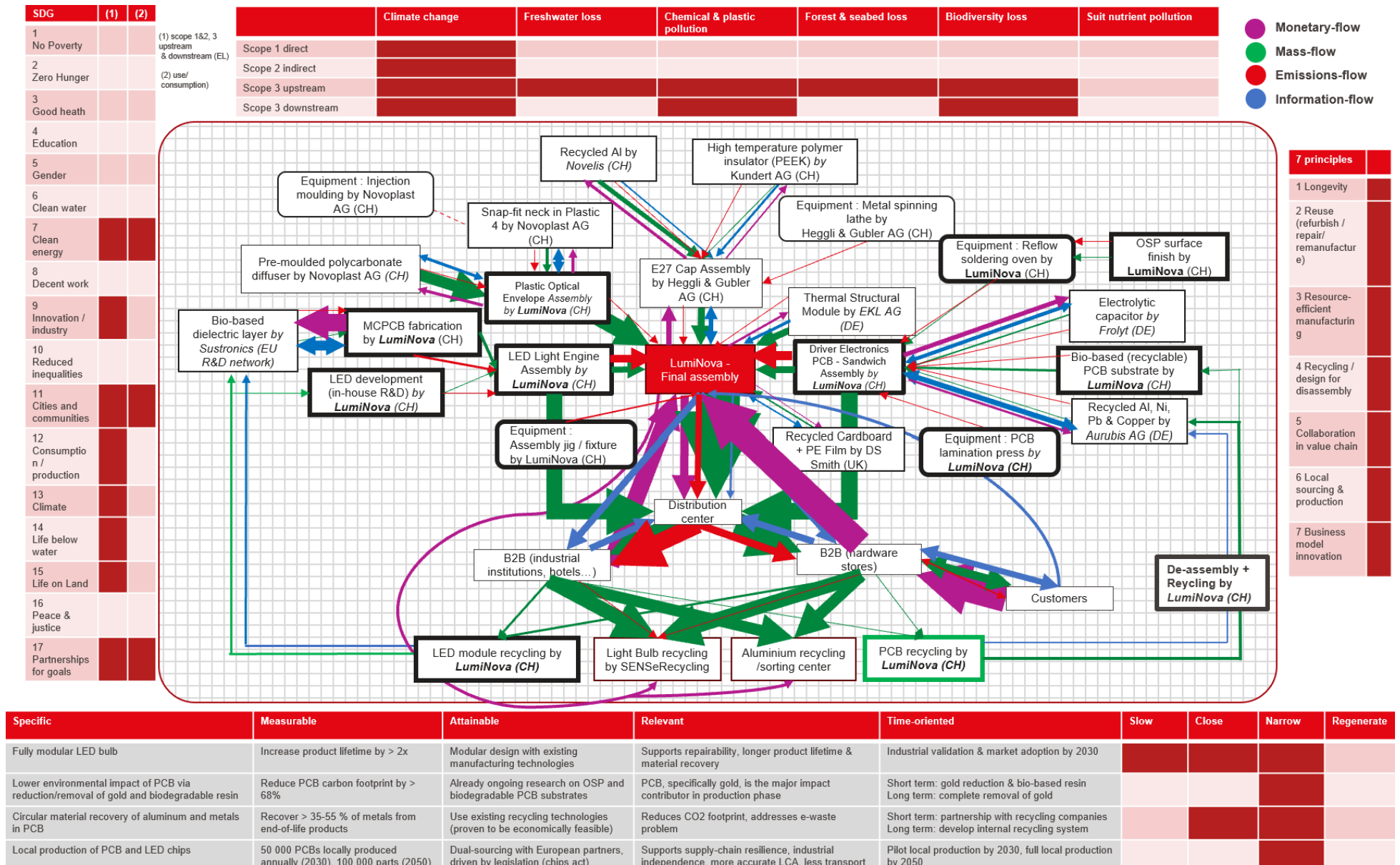


Figure S7: Circular Supply Chain - 2050

MFA of the light bulb Philips LED A60 E27 bulb, SKU 8720169191655

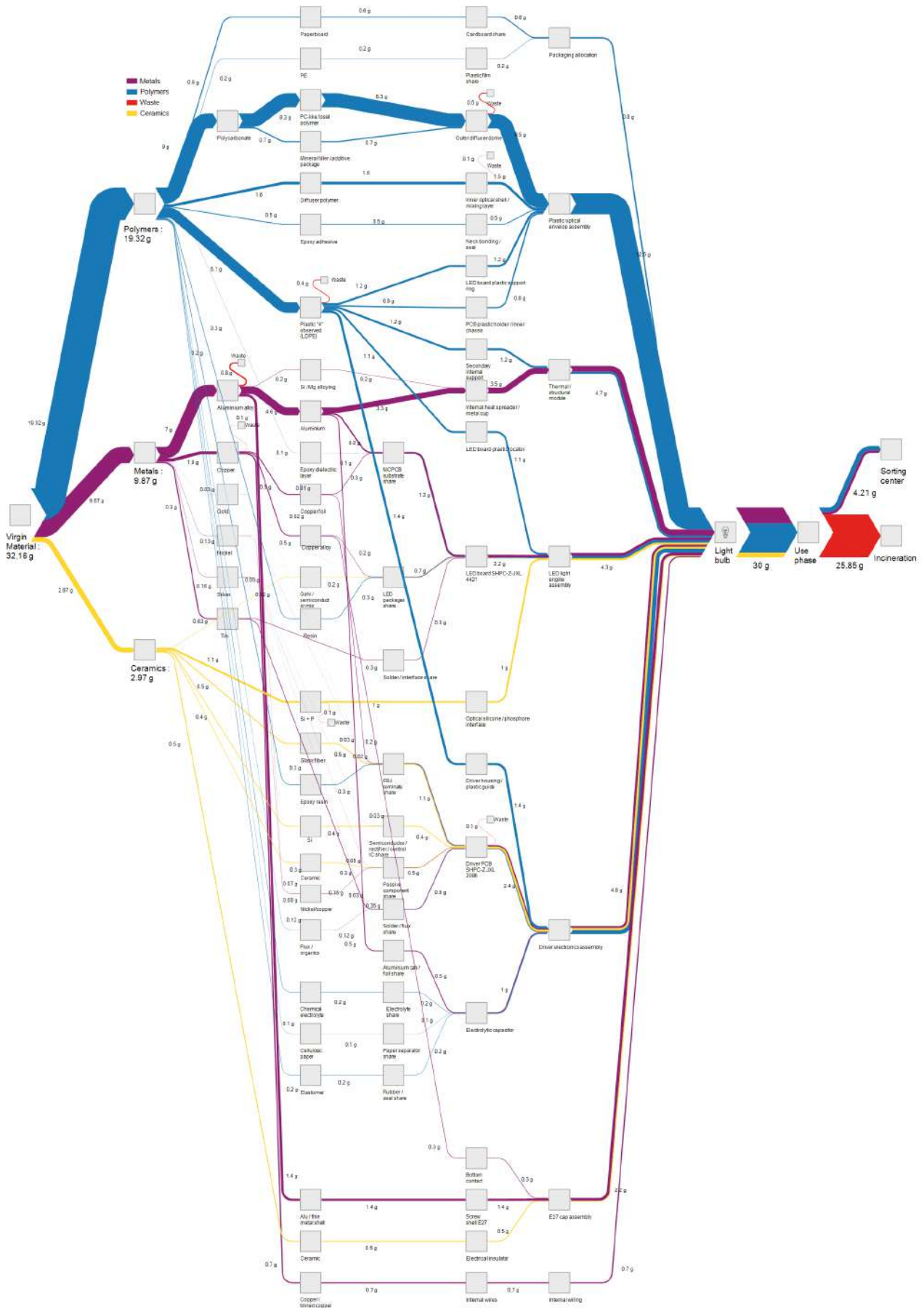


Figure S8: MFA of the linear incumbent

MFA of the light bulb Philips LED A60 E27 bulb, SKU 8720169191655

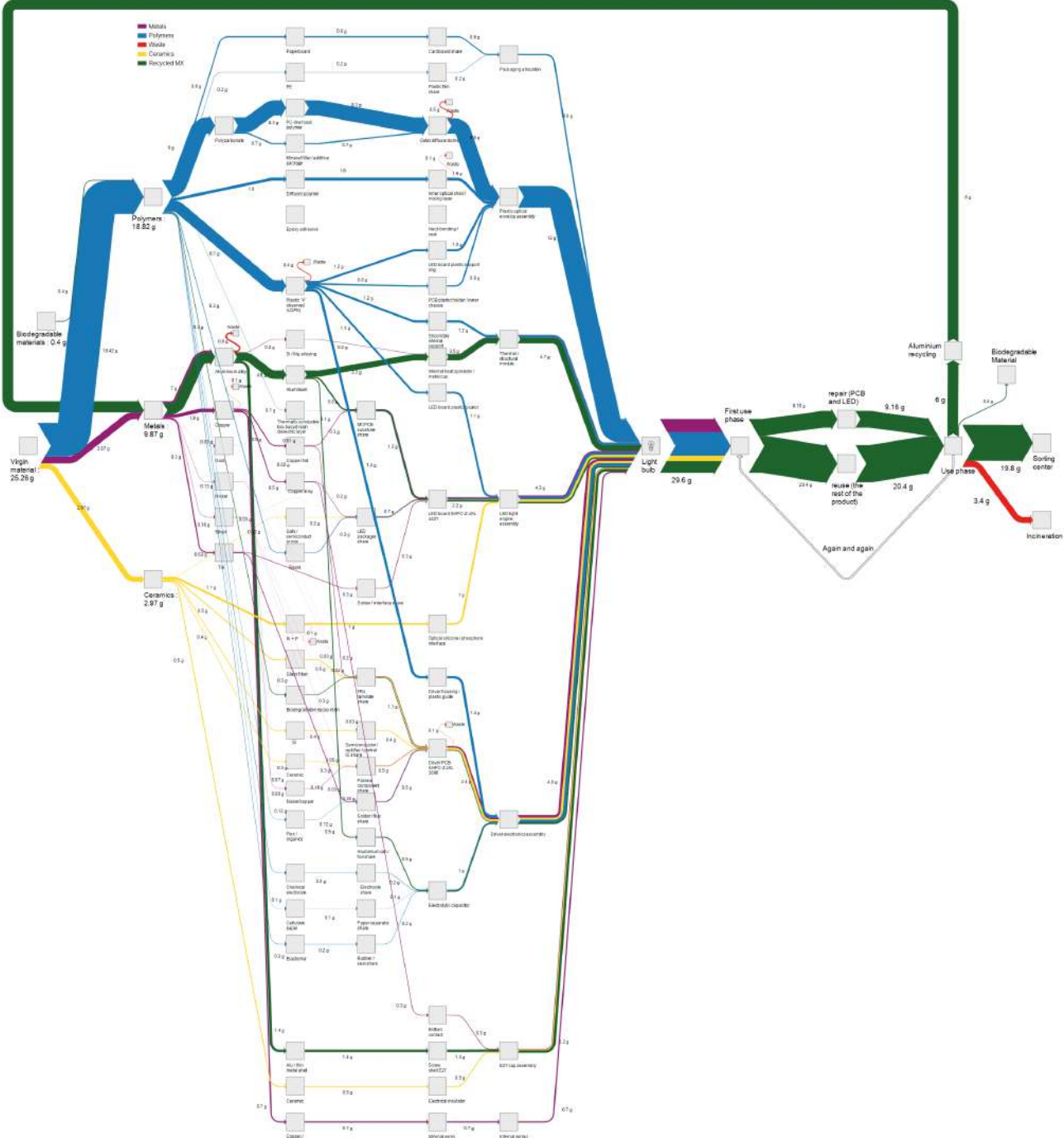


Figure S9: MFA of the new product

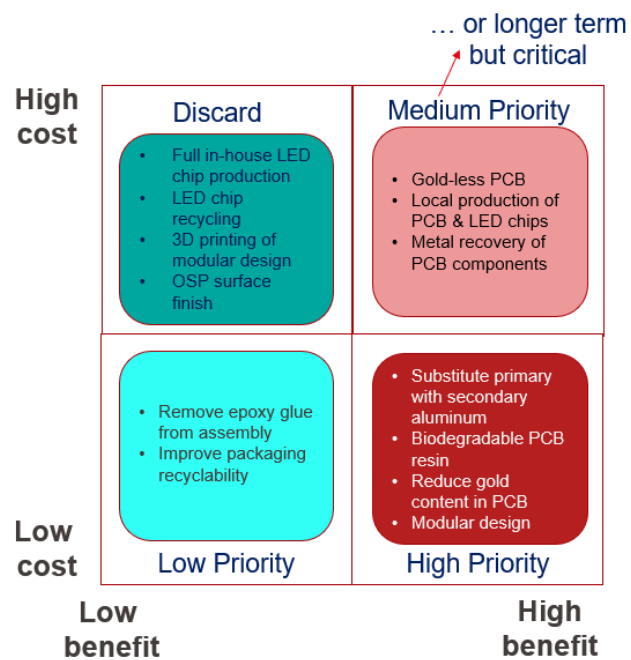


Figure S10: Brain-storming of initiatives

SMART objectives	1. Extend product lifetime through modular design	2. Reduce environmental footprint of PCB	3. Enable circular material recovery	4. Local production of PCB (2030) & LED chips (2050)
Specific	Design a fully modular LED bulb allowing the replacement of defective components (PCB / LED / driver) and elimination of epoxy (glue) seal	Develop and integrate an eco-designed PCB by: reducing gold content, alternative surface finish, biodegradable substrate, recycling of PCB metal contents	Implement a closed-loop recycling system for: PCB, aluminium, copper via partnerships with recyclers	Establish PCB sourcing, prototyping, and later full production in Switzerland
Measurable	Increase product lifetime by $\geq 2\times$ Reduce e-waste by $\geq 40\%$ per unit Reduce replacement frequency per household	Reduce PCB carbon footprint by $\geq 68\%$ compared to conventional FR4-based designs Reduce gold usage in PCB surface finish by $\geq 80\%$	Recover $\geq 35\%$ of metals from end-of-life products by 2030, targeting $\geq 55\%$ by 2050 as LED module recycling becomes feasible. Achieve $\geq 19\%$ recycled content in new products	By 2030 at least one PCB pilot line is qualified locally (about 50 000 parts produced) By 2050, full local production is reached (100 000 parts)
Achievable/with stretch	Achievable with current modular design and existing manufacturing technologies Stretch: strong user adoption and integration of an effective take-back program	Achievable via existing alternatives (ENIG substitutes) Stretch: integration of biodegradable PCB and complete elimination of gold	Achievable once metals are recovered with existing recycling technologies Stretch: efficient PCB material separation, reuse and valorization of recovered materials (e.g. reintegration into new PCBs)	Achievable via dual-sourcing with European partners first Stretch: move all PCB and LED chip fabrication and assembly locally by 2050
Relevant	Supports reparability (circularity principle), longer product life	PCB = major impact contributor (~5.52 kg CO ₂ eq) High leverage on sustainability	Addresses e-waste problem Aligns with circular business model (canvas) Medium term: establish partnerships with recycling companies (3-5 years)	Supports supply-chain resilience, shorter transport routes, and European industrial independence (e.g. Chips Act) Local production of PCB by 2030
Time-oriented	Short term: functional prototype (1–2 years) Medium term: industrial validation and market adoption (3–5 years)	Short term: gold reduction (1–2 years) Medium term: eco-PCB with biodegradable resin (3–5 years) Long term: complete gold removal (15-25 years)	Long term: develop internal recycling infrastructure to recover and reuse materials within the company (5+ years)	Full local production by 2050
Expected IMPACT	Significant reduction in electronic waste Lower frequency of full product replacement Promotion of a repair-oriented consumption model Better user trust	Lower CO ₂ footprint Reduced critical material dependency Improved recyclability	Reduced raw material extraction Lower environmental footprint Economic value from recovered materials	Stronger supply security, lower transport emissions, more control over quality and worker rights

Figure S11: SMART Initiatives

Initiative	Next steps	Change management barriers & challenges	Resource needs	Key dates
Modular, disassemblable bulb (N°1)	Define disassembly architecture, prototype and test repair/disassembly cycles, verify thermal and mechanical integrity Design a modular system which is simple (keep interface identical to a normal bulb), simple replacement instructions, develop a take-back/repair service and a failure detection system (identify component which fails)	Modular joints can weaken mechanical robustness, increase of failure points. Customer compliance/willingness (fear of difficulty of use or questioning of fragility, bulkiness, or hard to replace (perceived complexity))	Mechanical/thermal engineers, prototype tooling/labs, design for manufacturing, disassembly (DFD) specialists, marketing strategist, branding and packaging designers	Prototypes by 2027-2028, design locked for disassembly target by 2030
Biodegradable resin (N°2)	Find candidate substrates, run electrical, thermal, humidity, and aging tests. Test prototypes and optimize design. Partner with suppliers.	Needs to comply to electrical, thermal, flame-retardancy and long-term aging performance. Heat may accelerate failure.	Materials lab, PCB suppliers, compliance testing, testing budget,	2026–2027: material screening, 2028–2029: pilot boards, 2030: resin replacement target
Reduction + Recycling (2030) and complete removal (2050) of gold in PCB (N°2)	Map gold use by component, replace non-critical gold first, test alternative finishes and interconnects, revise supplier requirements, evaluate emerging gold recycling techniques	Premature failures due to poor contacts/interconnects and lower shelf-life of OSP substitutes	Engineers, reliability testing, corrosion testing	2026-2028: Gold inventory and testing of first substitutions, 2030: Recovery and reduction milestone, 2050: Complete removal target
Recycled aluminium (N°3)	Qualify recycled alloy streams, define impurity limits, test thermal/corrosion performance	Adequate thermal performance and corrosion resistance requires control of alloy composition and limited contamination	Metal supplier, materials quality assurance, thermal testing,	2026: supplier screening, 2027-2028: pilot, 2030: fully recycled content
Metal Recovery (N°3)	Implement a closed loop recycling system, evaluate emerging recovery methods	Economic feasibility and manufacturing compatibility (dis/assembly-line constraints)	Research leaders in metal recovery from PCBs, recycling companies, CAPEX for new disassembly line	2030: recovered and recycled Pb, Ni, Al, Cu and Au from PCBs
Local production of PCB and LED chips (N°4)	Qualify Swiss/EU fabrication plants and assemblers, start with prototypes at low volume, dual sourcing, expand to regional production capacity over time	Supplier qualification and scaling is non-trivial. May be limited by cost, capacity, and lead times with respect to Asian supply chains.	A stable industrial base and policy support. Supplier quality, capex/partnership funding, DFM engineering	2026-2027: supplier qualification, 2030: complete local production in-house of PCB, 2050: local production LED chips

Figure S12: Change management barriers & challenges

Stakeholder (Person/LinkedIn)	Organization	Job Title	Influence	How to contact	How to interact	Win-win, next steps	(Relational) barriers/opportunities
 Matt Sandberg Senior Researcher at RISE	RISE – Printed Electronics Area	Senior Researcher	High	LinkedIn, RISE Printed Electronics Arena contact channels, conferences	Approach with a precise research brief, request an expert discussion or collaboration meeting	RISE gains an industrial case study on recyclable PCBs, Lumivox gains expert input on recyclable resin systems and PCB design constraints.	Opportunity: strong technical credibility, direct relevance to recyclable PCBs, and potential access to research insights. Barrier: academic/research timelines may be slower than industry needs, and collaboration may require a clear scoping to fit their research focus.
 François Guber Managing Director at Circular Economy Switzerland	Circular Economy Switzerland (NGO)	Managing Director (CEO)	High	LinkedIn, website contact form, circular-economy events	Request an introductory meeting or short advisory call	Project becomes a visible Swiss circular-economy case study, strengthening the national circularity narrative and giving Circular Economy Switzerland a concrete example of local, repairable, future-oriented manufacturing	Opportunity: strong access to Swiss circular-economy network and visibility. Barrier: limited technical expertise for product-specific decisions, would need to secure modular design first
 Carlo Giordano Sustainability Lead and Leadership Senior Advisor at Deloitte	Deloitte	Sustainability Lead and Leadership Senior Advisor	Medium	Email	Mention our meeting previously and his insightful talk on metal recovery from batteries. Request a quick meeting	For Carib/Deloitte, the project is a concrete case study in circular product innovation and sustainability transformation. Carlo had previously expressed interest in recovering PCB materials. For Lumivox, access to strategic framing, impact assessment, and business model insight	Opportunity: high-level expertise in sustainability transformation, circularity, and stakeholder facilitation. Barrier: Need to secure technical feasibility first
 David H. Spritz Lead Consulting Engineer, DFM engineering expert, 30 years experience in PCBA manufacturing at Ansys	Ansys	Lead Consulting Engineer, DFM engineering expert, 30 years experience in PCBA manufacturing	High	Professional email, conference networking	Request an expert advisory call or short technical review	Ask for guidance on gold-less PCB design, solderability, SMT compatibility, DFM constraints, and feasibility of local PCB production	Opportunity: strong expertise in DFM/SMT as well as PCB surface finishes Barrier: may need to adapt advice to Switzerland and LED-bulb constraints
 Anis Nassar Head of Circular Economy Innovation and Business Engagement, Head in Critical Minerals at World Economic Forum	World Economic Forum	Lead in Circular Economy Innovation and Business Engagement, Head in Critical Minerals	Medium	WEF profile, LinkedIn, WEF events, introductions through circular-economy networks	Approach with a short concept note focused on circular value, resource efficiency	Ask for feedback on business adoption, behavioral levers, and relevant WEF initiatives. Would allow for network visibility and policy/business insight	Opportunity: directly aligned with his circular-economy portfolio and local to Switzerland/Fribourg Barrier: WEF is high-level and network-oriented, may not be as straightforward to collaborate for a pilot project
 Petr Vesely Assistant Professor (Department of Electrotechnology), Leads the JOP Lab at Czech Technical University in Prague	Czech Technical University in Prague - Faculty of Electrical Engineering	Assistant Professor (Department of Electrotechnology), Leads the JOP Lab	Medium to high	Email, CTU faculty page, research page	Send a concise academic outreach note with our concept and technical questions	For Prof Vesely, real-world application opportunity for their research in biodegradable substrates for circuit boards. For Lumivox, expert feedback in materials and manufacturability	Opportunity: high technical relevance and interest in sustainable electronics. Barrier: may not be able to translate academic success into industrial application
 Tighaine Treise CEO of Temeloy and helped launch Lighting for Good	Lighting for Good (non-profit) and Temeloy	CEO of Temeloy and helped launch Lighting for Good	Medium	Website Contact page, LinkedIn	Contact with a one-page concept note and ask for a conversation	Help validate the modular concept and create a pathway toward pilot market introduction of the product Ask for eco-design visibility and sustainability networking For LFG: Bulb becomes a reference project for circular lighting design, showing how products can be designed for disassembly, repair, and reduced environmental impact	Opportunity: good mission fit, could help build industry credibility Barrier: Established in London, may lack local expertise
 Matthias Baeth Head of Raw material cycles, deputy head of the Waste and Resources Division at Federal Office for the Environment (FOEN/BAFU)	Federal Office for the Environment (FOEN/BAFU), specifically sector Circular Economy and Ambient Pollution Level	Head of Raw material cycles, deputy head of the Waste and Resources Division	High	FOEN/BAFU website contact channels, policy consultations and federal sustainability events	Approach with a concise policy-relevant project brief, request a short meeting or written feedback; frame the project as a circular-economy and resource-efficiency case	For FOEN/BAFU the project provides a concrete example of repairable, recyclable, locally sourced electronics aligned with Switzerland's circular-economy objectives. For Lumivox: access to policy guidance, legitimacy, and potential alignment with future regulatory developments	Opportunity: high policy relevance, strong legitimacy, and alignment with Swiss circular-economy priorities. Barrier: the office is policy-oriented; engagement may require a clear public-interest framing
 Attila Cécely Associate Professor in Faculty of Electrical Engineering and Informatics at Budapest University of Technology and Economics	Budapest University of Technology and Economics	Associate Professor in Faculty of Electrical Engineering and Informatics	High	Faculty/research page, email, conference networks	Concise academic outreach mentioning his work on biodegradable substrates, circular PCB design, and WEEE recovery, then ask for a short technical discussion	For him: project could be a real-world case for his biodegradable PCB. For Lumivox: access to expertise on substrate materials, PCB manufacturability and metal recovery	Opportunity: group already demonstrated working biobased PCB substrates, circular PCB design rules, and bioleaching for critical metal recovery Barrier: likely research-focused, so collaboration is not straightforward
 Andreas Bili Project Manager and Scientist for circular economy and waste management projects at Empa	Volube and Empa	Sustainability Consultant at Volube Project Manager and Scientist for circular economy and waste management projects at Empa	Medium to high	Volube website, Empa website, professional email, LinkedIn	Pitch the project briefly, mention the EPFL connection, and ask for his expertise on recycling system design, compliance, and take-back feasibility	For Andreas: project provides a concrete case for evaluating how recycling and take-back systems could support circular electronics. For Lumivox: advice on compliance, system design, monitoring, and feasibility of recovery pathways	Opportunity: strong expertise in e-waste recycling, audits, and Swiss waste-system standards. Barrier: would need to validate technical feasibility of metal recovery first
 Raffaele Mezenga Professor in Department of Health Sciences and Technology at ETH Zurich	ETH Zurich	Professor in Department of Health Sciences and Technology	High	Faculty/research page, email, conference networks	Approach in a concise academic style, with a clear question about e-waste gold recovery and scale-up potential	For him: project could be a practical case for circular electronics. For Lumivox: access to a gold recovery method with strong evidence of technical and economic feasibility	Opportunity: strong credibility and direct relevance to metal recovery initiative. Barrier: work is highly research-driven, so likely best for scientific validation and strategic insight, may not be industrially applicable

Figure S13: Stakeholders to Engage

Net Present Value (NPV) Analysis

The financial viability of the LumiNova plant was assessed with the course NPV/SROI tool over a seven-year horizon. The base case assumes a ramp-up trajectory of 100 000 bulbs in Year 1, 500 000 in Year 2, and 1 000 000 bulbs/yr from Year 3 onwards, sold at a unit price of 9.99 €. A weighted average cost of capital (WACC) of 9%, a corporate tax rate of 14.6% (representative of the Swiss assembly location), and the full ReCiPe 2016 central environmental shadow cost of 4.03 €/part (derived from the endpoint LCA monetisation in Section 3.5.3 using CE Delft central prices (*Bijleveld et al., 2024*)) are applied. The capital expenditure of 3.00 M€ (equipment + tooling) plus a front-loaded R&D capital of 0.45 M€ are incurred in Year 0, for a total initial outlay of 3.45 M€ consistent with the full-scale project CAPEX of Table 5. The key assumptions and outputs are summarised in Tables S1 and S2.

Table S1: NPV model assumptions and headline results.

Parameter	Value
Project horizon	7 years
Selling price	9.99 €/bulb
WACC (discount rate)	9.0 %
Corporate tax rate	14.6 %
CAPEX (equipment + tooling)	3 000 000 €
R&D CAPEX (Year 0)	450 000 €
Environmental shadow cost (full ReCiPe central)	4.03 €/part
Total parts over horizon	5 600 000
Net Present Value (NPV)	13.53 M€
Internal Rate of Return (IRR)	58.7 %
Payback period	2.43 years
Accounting ROI	45.8 %

Table S2: Discounted cash-flow table (WACC = 9%). All values in €; cash flows assumed at end of period.

Year	Volume	Revenue	Cash flow	PV	Cumulative PV
0	–	–	–3 450 000	–3 450 000	–3 450 000
1	100 000	999 000	–159 600	–146 422	–3 596 422
2	500 000	4 995 000	1 514 996	1 275 142	–2 321 280
3	1 000 000	9 990 000	4 841 326	3 738 392	1 417 112
4	1 000 000	9 990 000	4 841 326	3 429 717	4 846 829
5	1 000 000	9 990 000	4 841 326	3 146 530	7 993 359
6	1 000 000	9 990 000	4 841 326	2 886 725	10 880 083
7	1 000 000	9 990 000	4 841 326	2 648 371	13 528 455

The project breaks even on a discounted basis during Year 3, when the cumulative present value turns positive. The strong NPV (13.53 M€) and IRR (58.7%) confirm that the circular LumiNova bulb is financially attractive even when the environmental shadow cost of 4.03 €/part dominated by human-health damages (88.5%) as identified in Section 3.5.3 is taken into account. This provides the economic foundation for the social-value analysis below.

Social Return on Investment (SROI) Analysis

The SROI framework compares the discounted social, environmental and economic value created with the total investment required. Following the course tool, the numerator combines three streams: (i) the economic *value-in-use* delivered to customers, since one modular LumiNova bulb provides 60 000 h of lighting service and therefore replaces four conventional bulbs (value-in-use = $4 \times 9.99 \text{ €} = 39.96 \text{ €/bulb}$); (ii) the avoided environmental externalities between the incumbent Philips bulb and LumiNova, monetised from the ReCiPe 2016

LCA results with CE Delft central prices (Bijleveld et al., 2024) (the incumbent shadow cost of 5.80€/part is estimated by scaling LumiNova’s 4.03€/part by the ratio of total LCA climate impacts 7.21/5.01, giving 1.77€/part avoided); and (iii) the direct stakeholder reinvestment drawn from the three supply-chain partnerships of Section 3.5.5. Following standard SROI practice (SROI Network, 2012), the environmental benefits are corrected for deadweight (25 %), attribution (40 %), displacement (5 %) and an annual drop-off (10 %), giving a combined adjustment factor of 0.4275 in Year 1. A social discount rate of 3.5 % is applied over the seven-year horizon. The inputs and results are reported in Tables S3 and S4.

Table S3: SROI stakeholder reinvestment streams (per project, over the full lifetime), drawn from the NetPositive partnerships of Section 3.5.5.

Value stream	Basis	Amount
Closed-loop aluminium (Novelis)	100 000 € (2027) + 50 000 €/yr (2028–2030)	250 000 €
Sustainable PCB R&D (AT&S)	joint R&D programme 2027–2029	250 000 €
Take-back network (retailers)	150 000 € (2027) + 20 000 €/yr (2028–2033)	270 000 €
Total stakeholder reinvestment		770 000 €
<i>SROI correction factors (applied to environmental benefits)</i>		
Deadweight	share occurring anyway	25 %
Attribution	caused by other actors	40 %
Displacement	cancelling benefits elsewhere	5 %
Drop-off	reduction per year after Year 1	10 %

Table S4: SROI results (social discount rate = 3.5 %, full ReCiPe 2016 central pricing).

Metric	Value
PV of value-in-use (economic benefit)	190.94 M€
PV of avoided environmental externalities (corrected)	2.52 M€
PV of stakeholder reinvestment	0.71 M€
PV of total social, environmental & economic benefits	194.17 M€
PV of total investment (CAPEX + OPEX)	24.44 M€
Negative externalities to offset (project)	22.57 M€
Gross SROI ratio	7.95
Net SROI ratio (gross – 1)	6.95

A gross SROI of 7.95 indicates that every euro invested generates approximately 7.95€ of combined economic, social and environmental value. This value is driven primarily by the economic value-in-use of the modular architecture a single LumiNova bulb replacing four conventional bulbs over the 60 000 h functional unit complemented by the avoided human-health and ecotoxicity externalities monetised from the ReCiPe 2016 LCA. With the corrected shadow cost of 4.03€/part, the absolute negative externalities of the project fall to 22.57 M€ (from 60.48 M€ when an erroneous 10.76€/part was used), reducing the NetPositive gap by 63 %. The project is therefore not yet formally NetPositive the residual 22.57 M€ of externalities still exceed the 0.77 M€ of stakeholder reinvestment but the very high SROI shows that reinvesting a growing fraction of the avoided-externality value into the three supply-chain stakeholders progressively closes this gap, consistent with the absolute sustainability targets.